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(54) **ELECTRONIC DEVICE**

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(57)

ABSTRACT

An electronic device includes a first device and a second device. The first device and the second device are removably attachable to each other. The first device includes a first front surface, and a first display and a first connection portion arranged at the first front surface. The second device includes a second front surface, a second display arranged at the second front surface, a second rear surface located opposite to the second front surface, and a second connection portion arranged at the second rear surface. The second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction. In a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display.

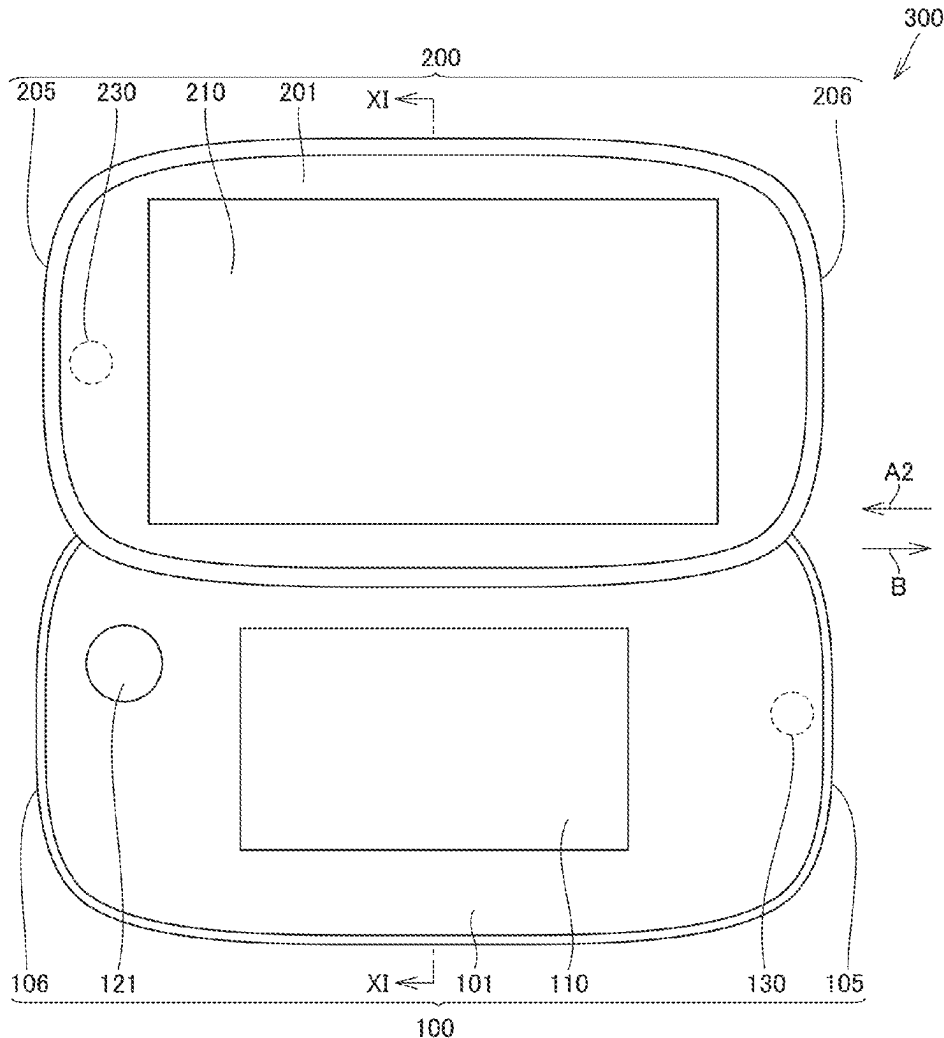


FIG.1

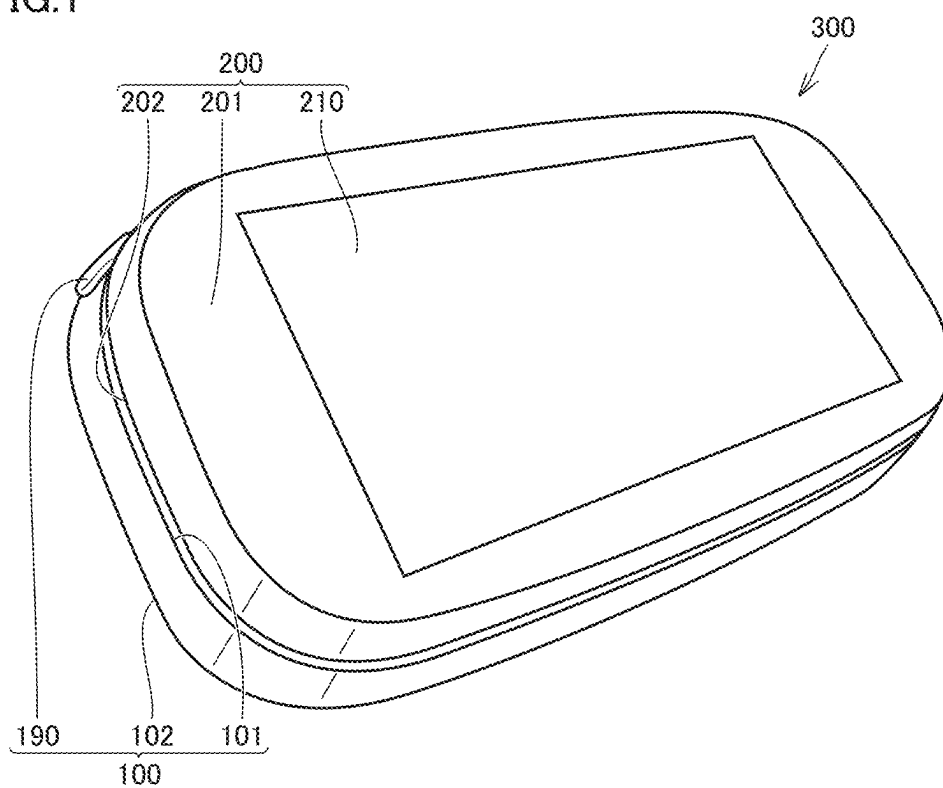


FIG.2

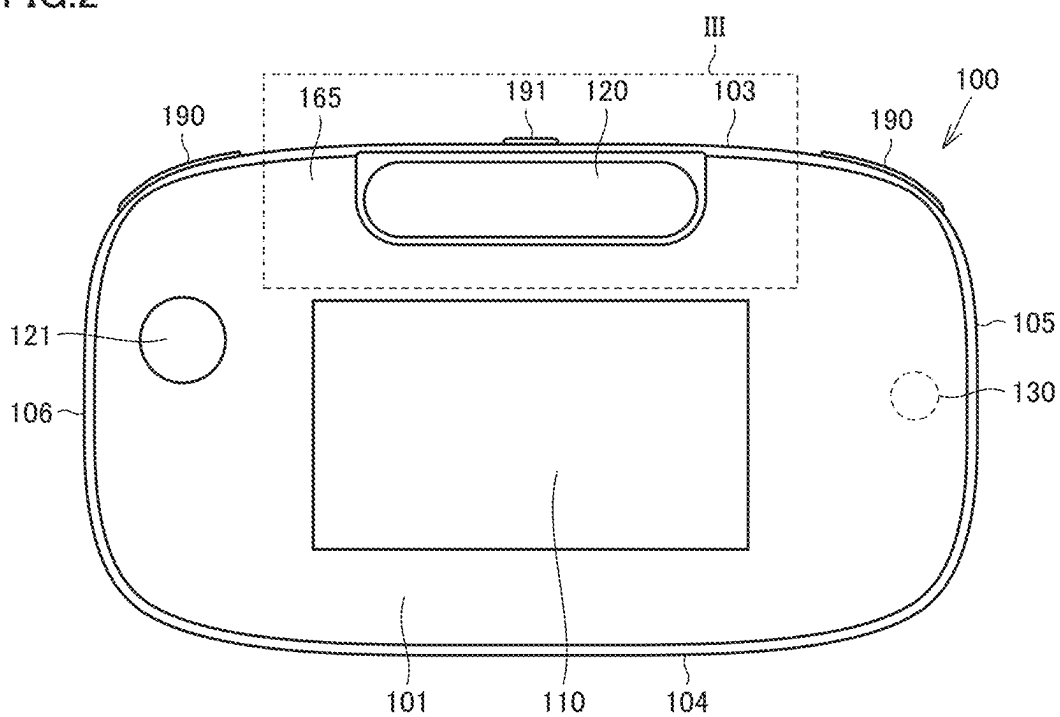


FIG.3

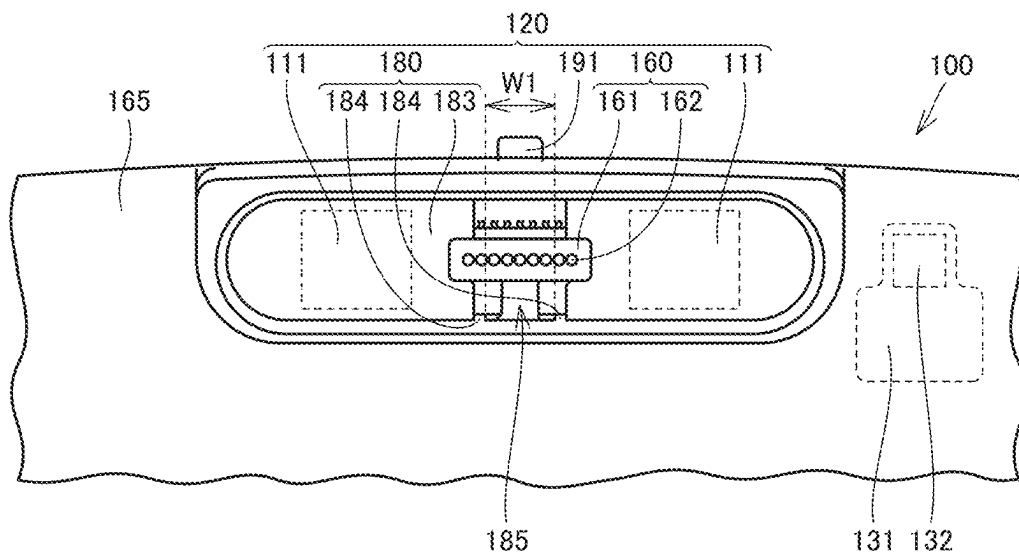
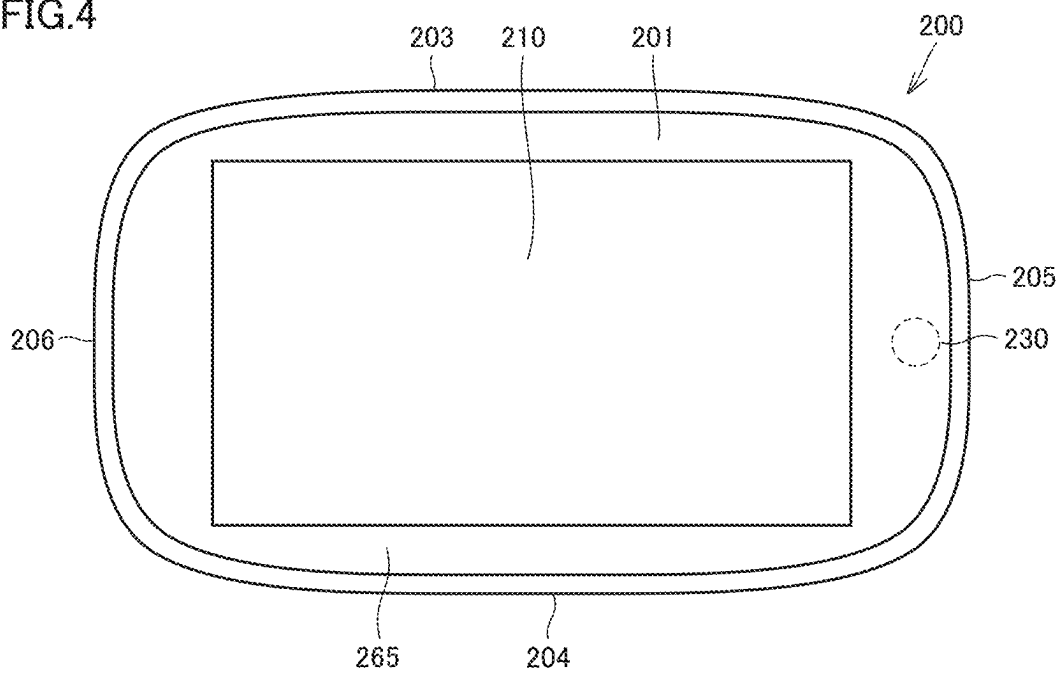


FIG.4



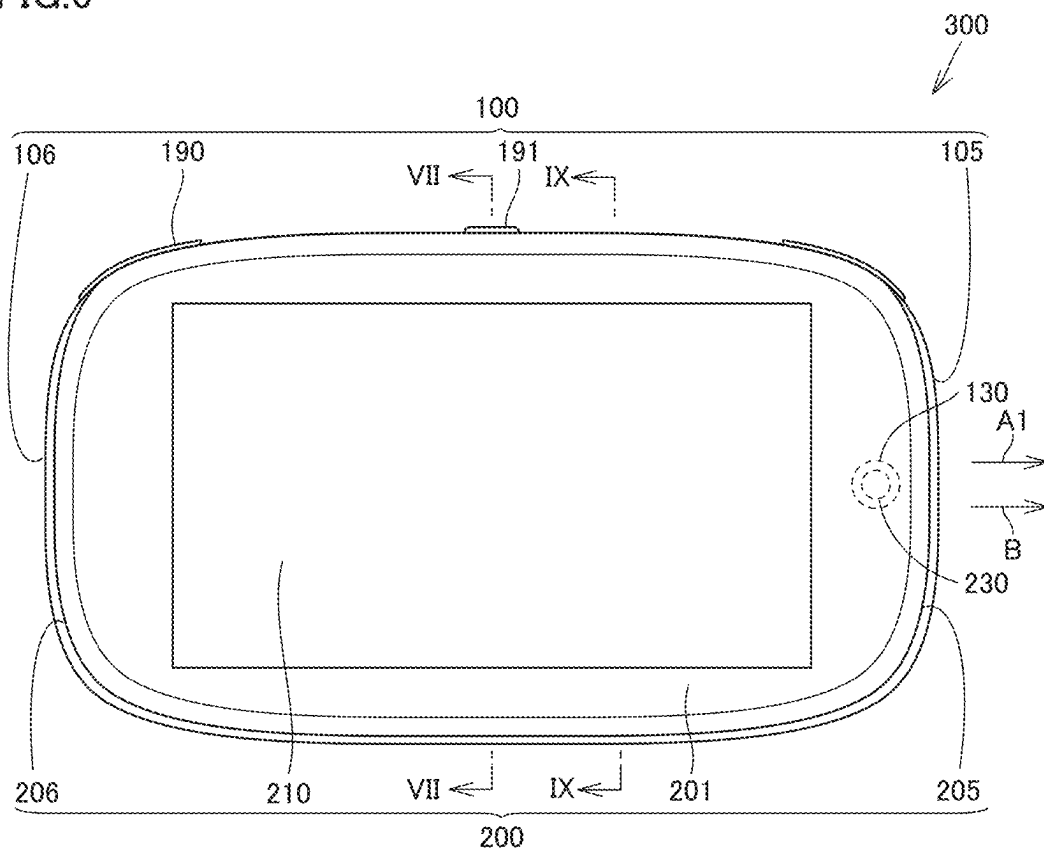


FIG.7

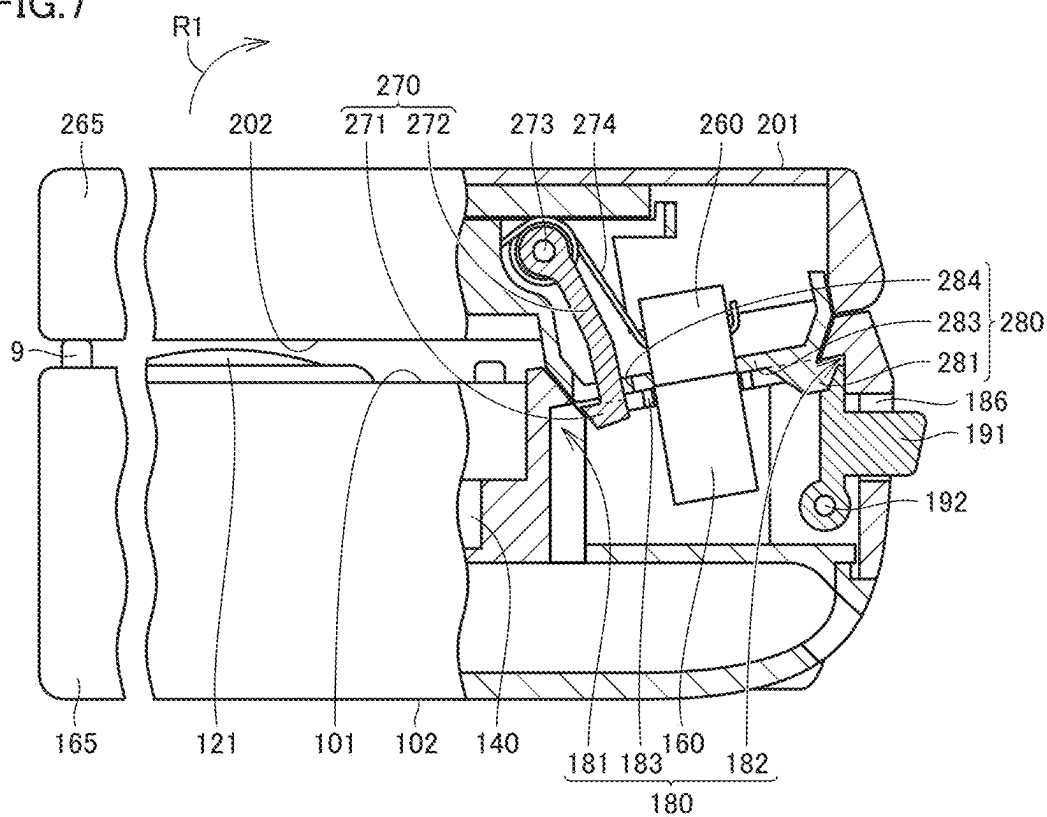


FIG.8

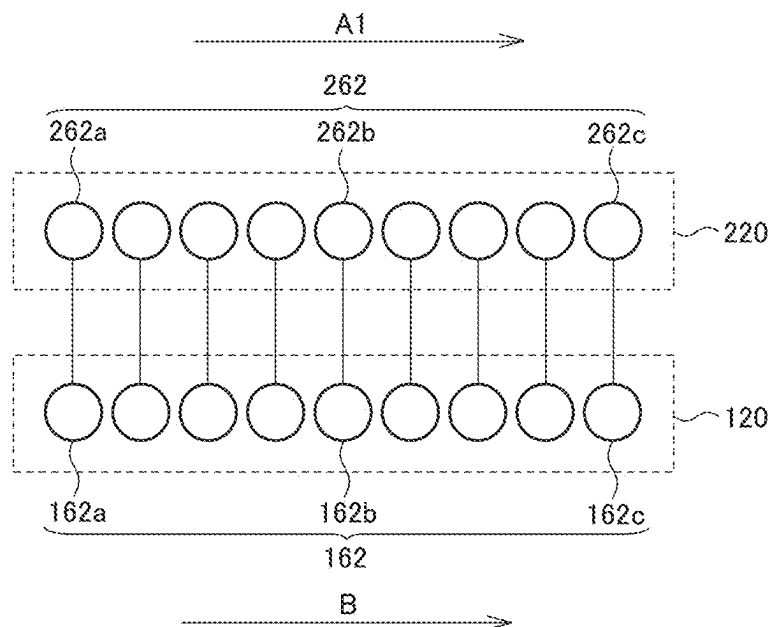
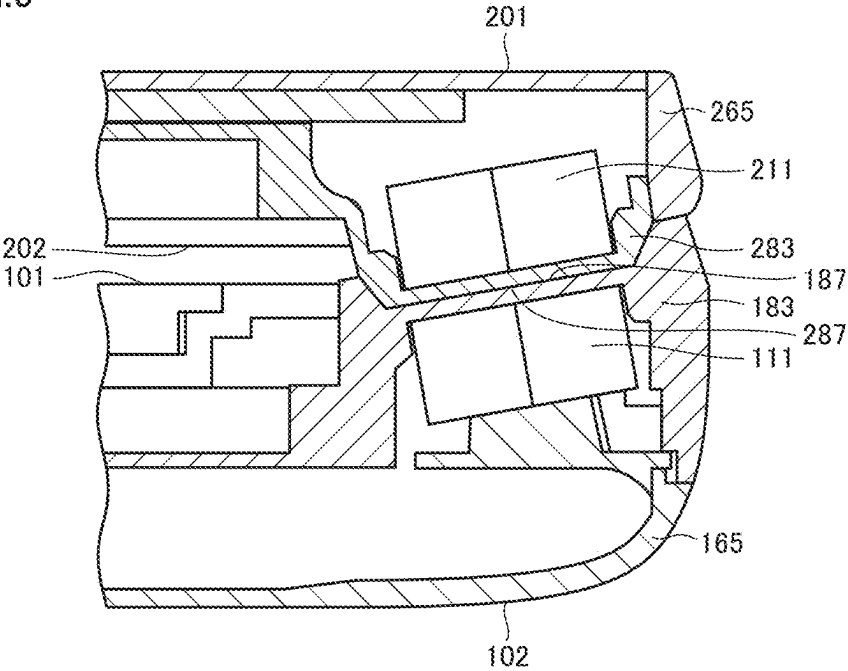


FIG.9



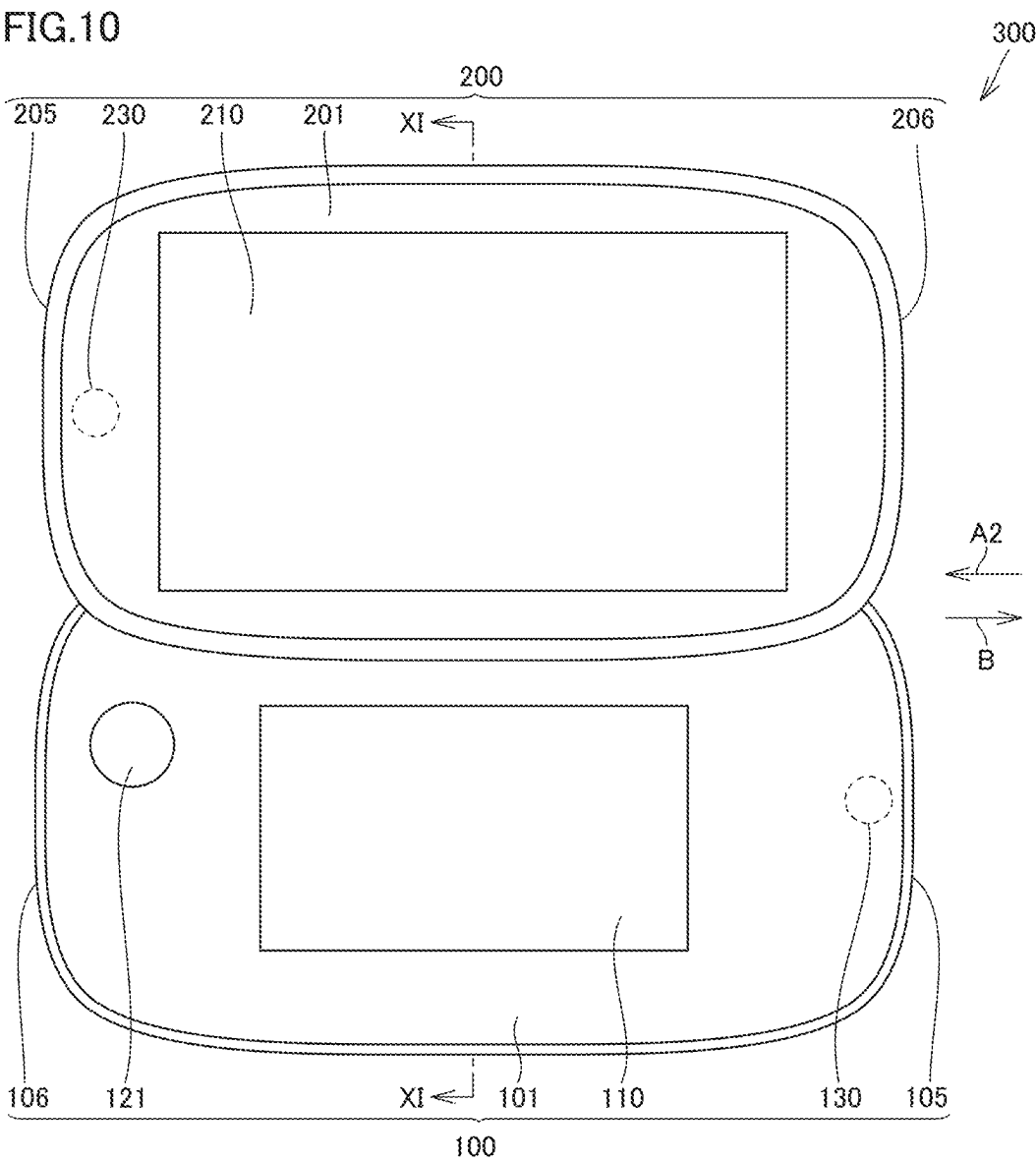
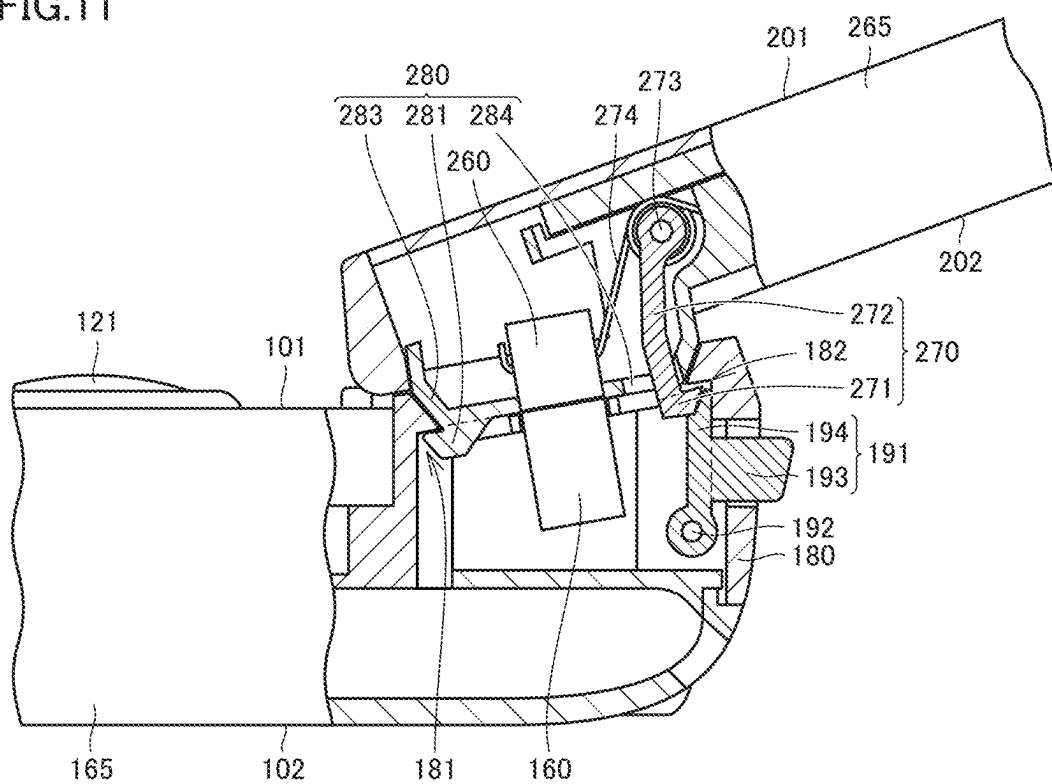


FIG.11



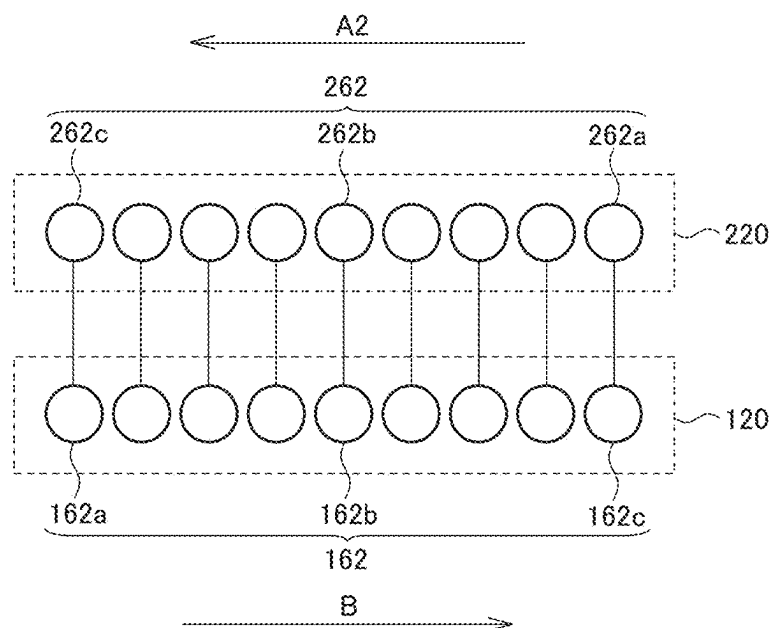


FIG.14

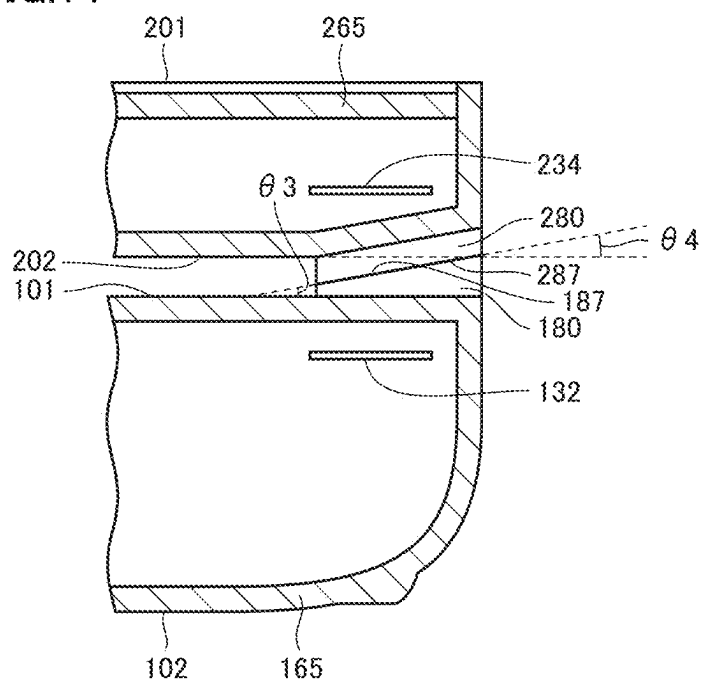


FIG.15

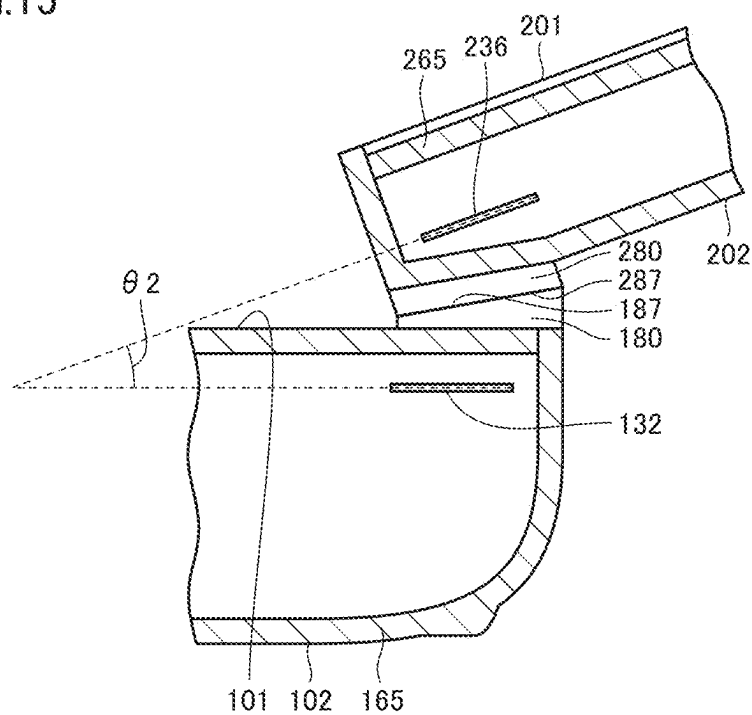


FIG.16

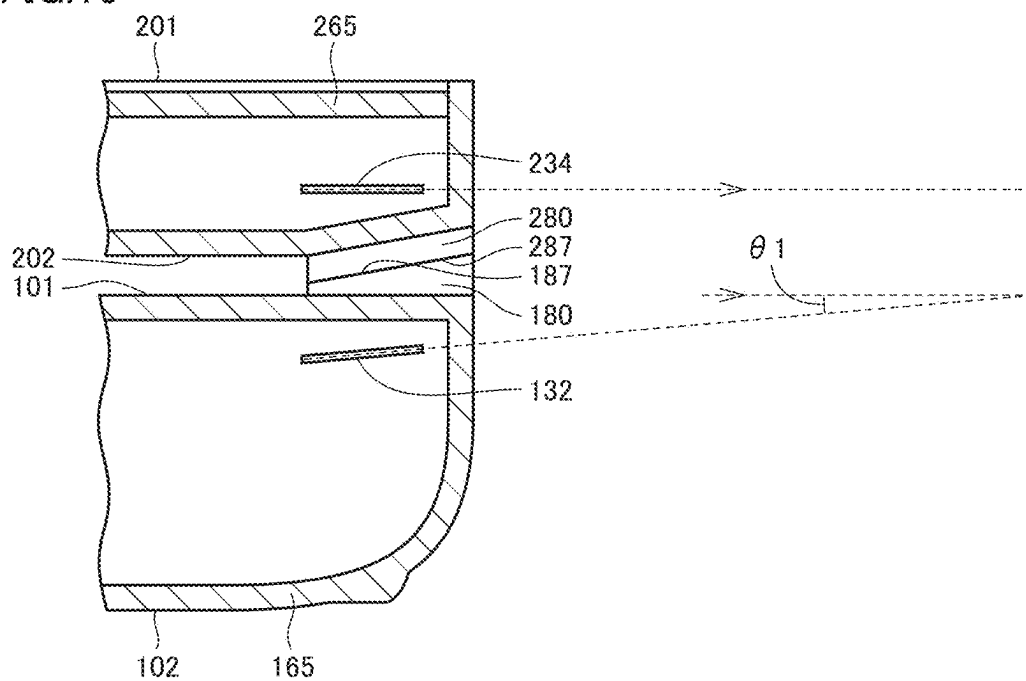
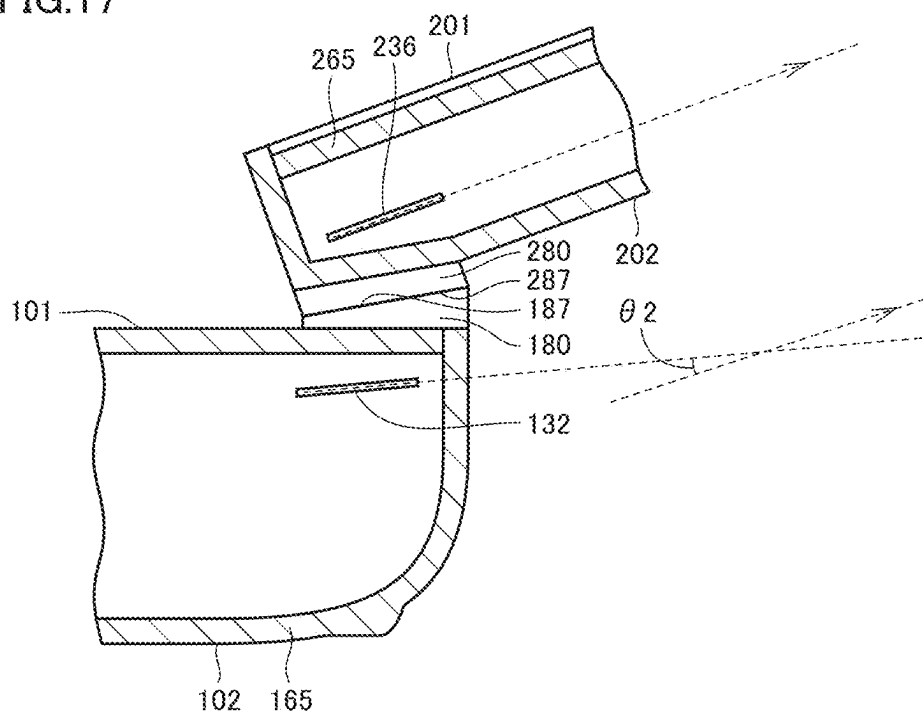


FIG.17



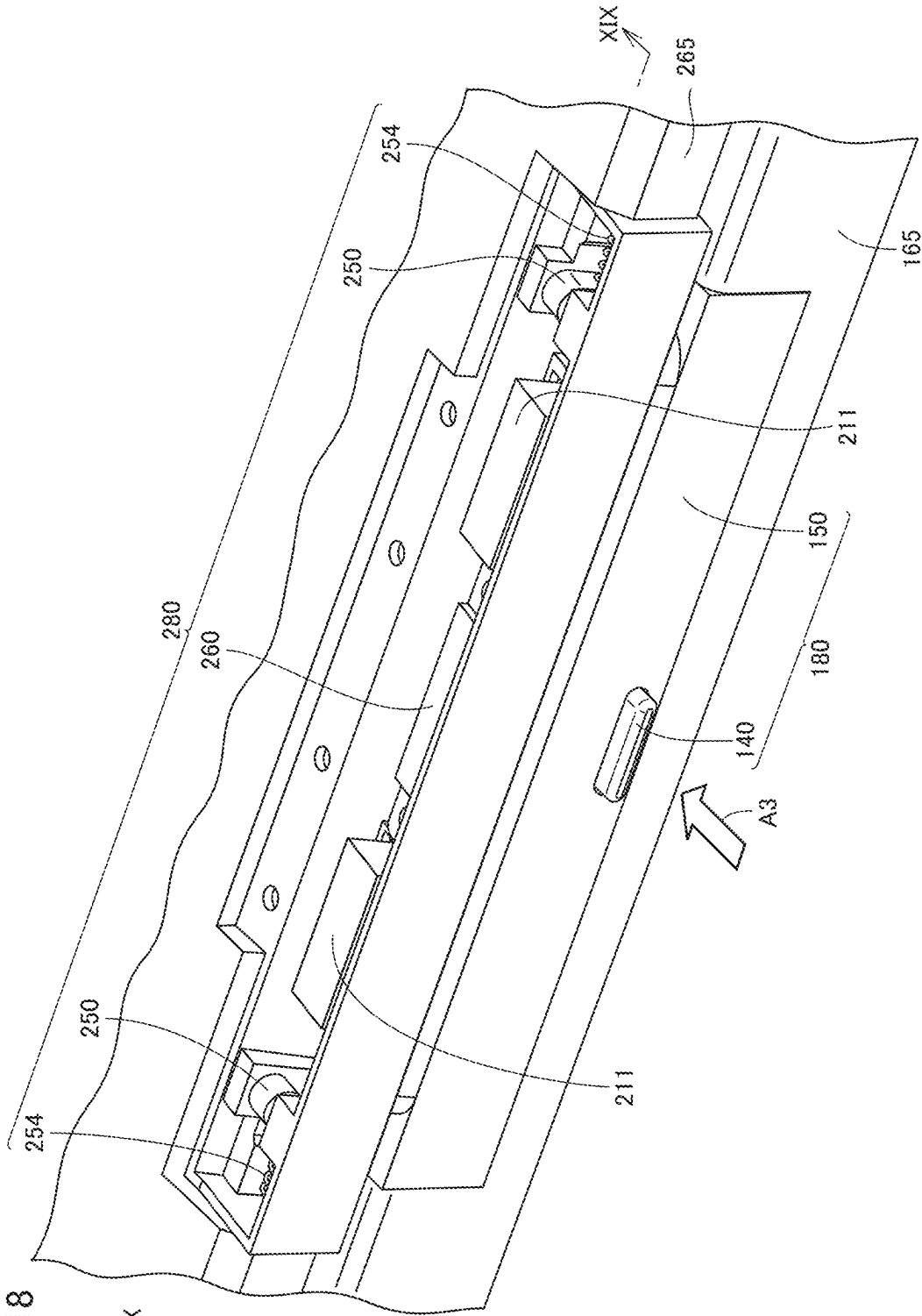


FIG. 18

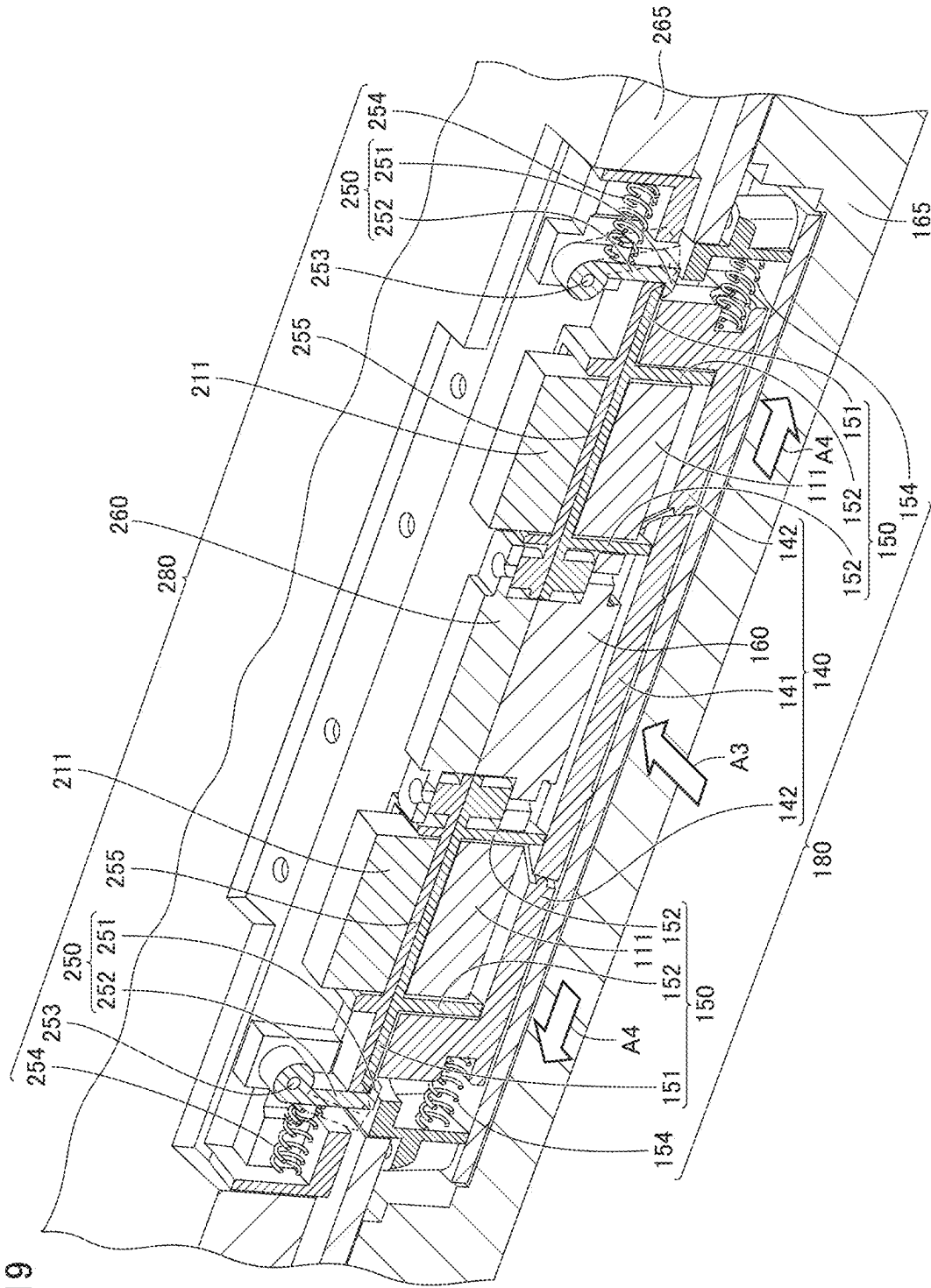


FIG.19

FIG.20

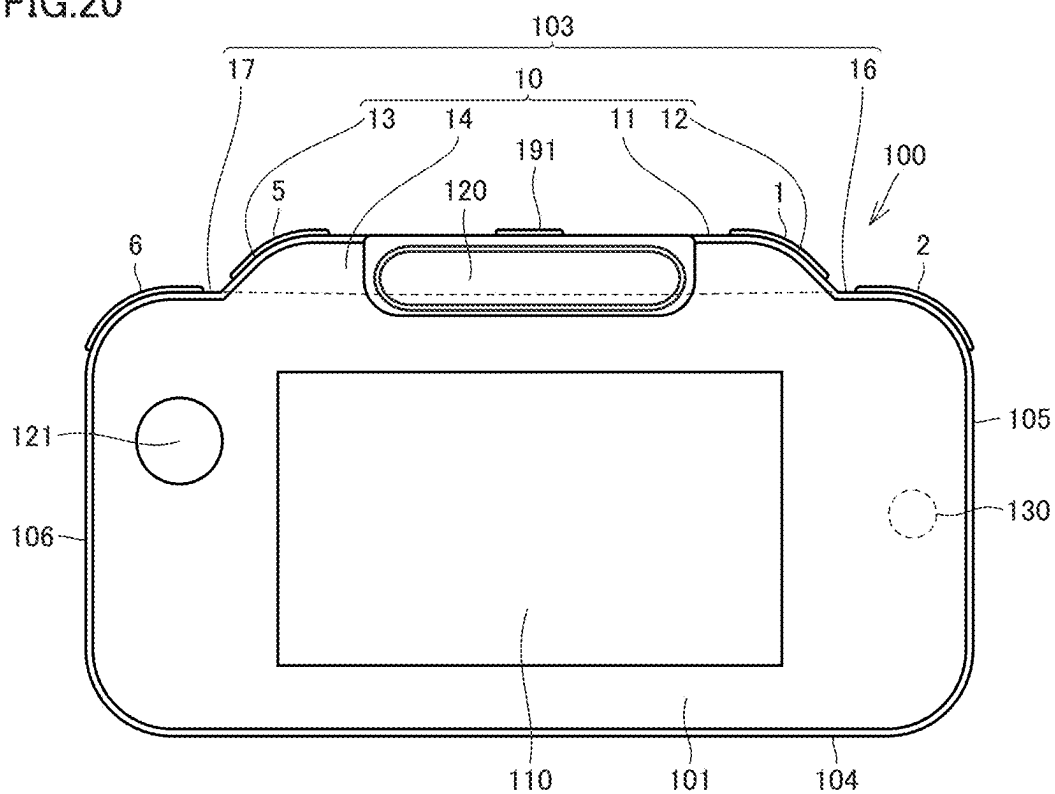


FIG.21

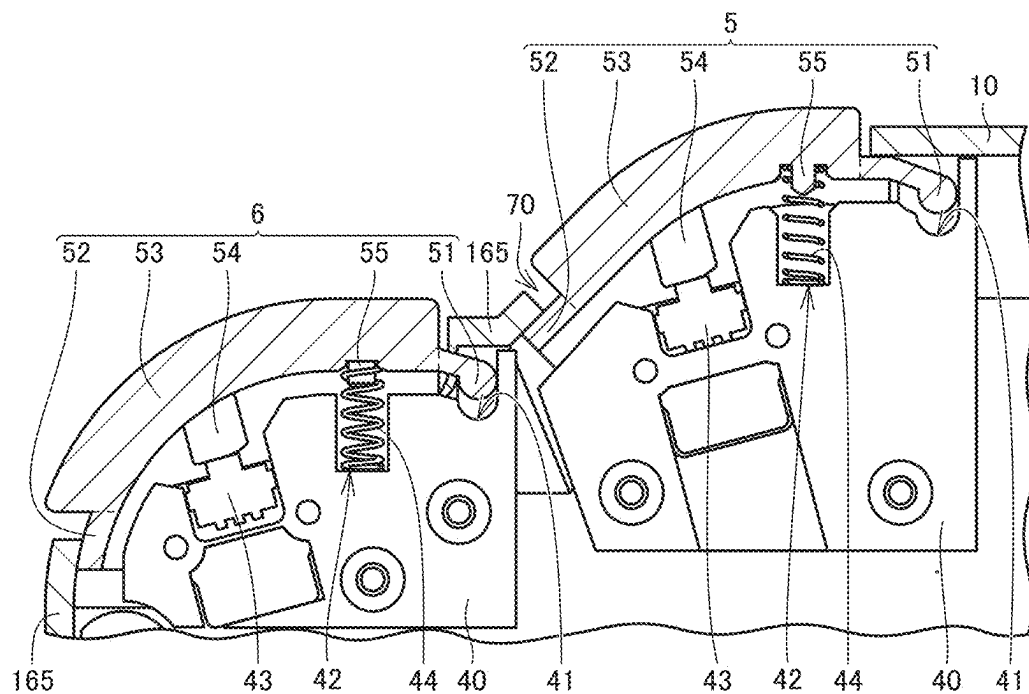


FIG.22

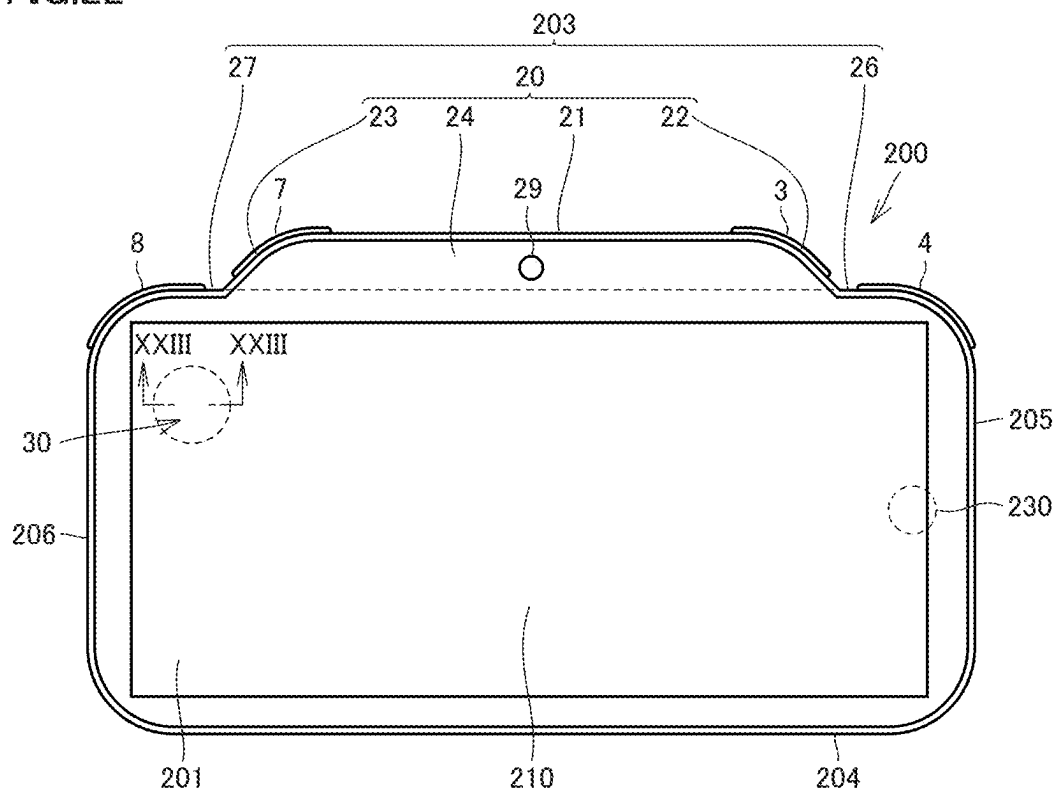


FIG.23

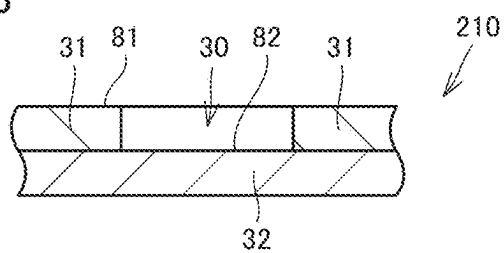


FIG.24

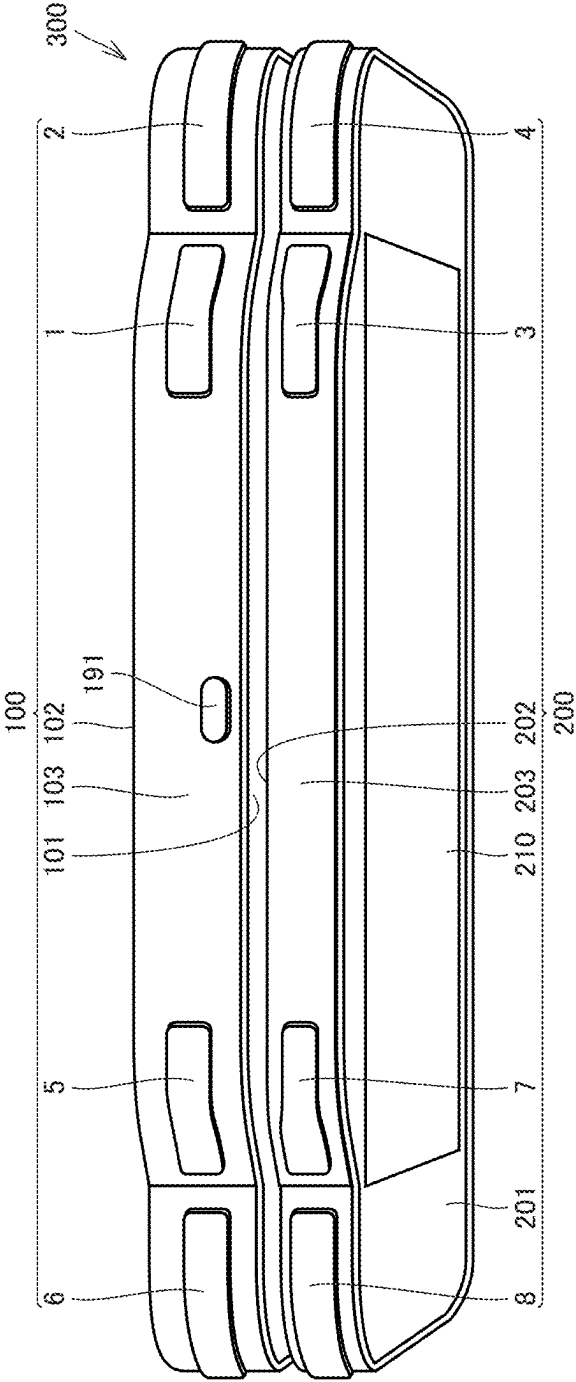
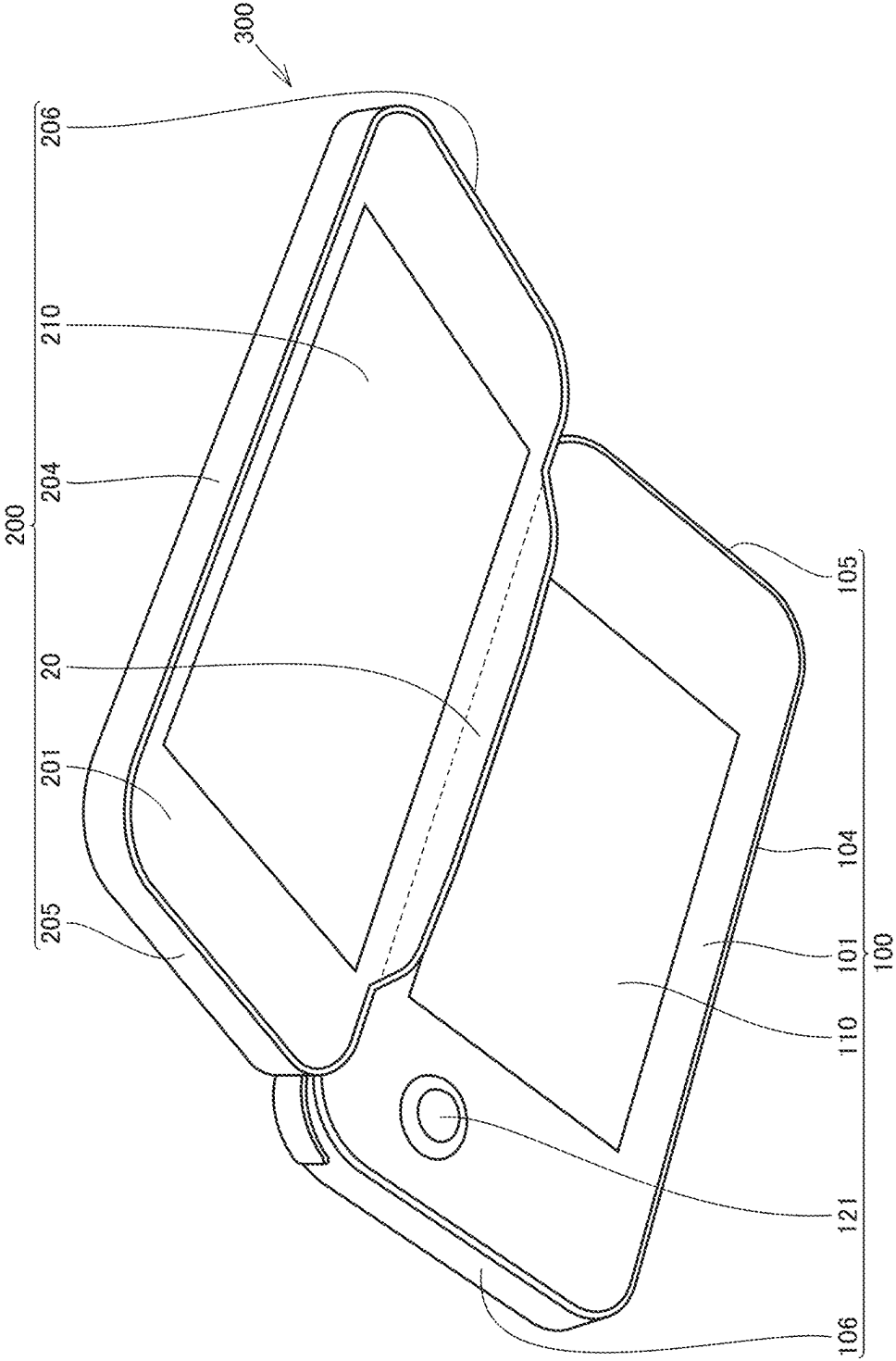


FIG.25



ELECTRONIC DEVICE

[0001] This nonprovisional application claims priority on and is a continuation of International Patent Application PCT/JP2022/042086 filed with the Japan Patent Office on Nov. 11, 2022, the entire contents of which are hereby incorporated by reference.

FIELD

[0002] The present disclosure relates to an electronic device.

BACKGROUND AND SUMMARY

[0003] A dual display mobile device has been known.

[0004] An exemplary embodiment provides an electronic device that includes a first device and a second device. The first device and the second device are removably attachable to each other. The first device includes a first front surface, and a first display and a first connection portion arranged at the first front surface. The second device includes a second front surface, a second display arranged at the second front surface, a second rear surface located opposite to the second front surface, and a second connection portion arranged at the second rear surface. The second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction. In a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display. In a second connection state in which the second connection portion is connected to the first connection portion in the second direction, the second rear surface is arranged not to cover the first display. Thus, a novel structure that switches between connection states is provided.

[0005] For example, in communication between the first device and the second device, an external terminal may be provided in each of the first device and the second device and the external terminals may be connected to each other. When the external terminal is provided in each of the first device and the second device, however, high-frequency parasitic emission may occur.

[0006] According to the electronic device according to the above, the first device may include a first substrate arranged inside the first device and one first antenna or at least two first antennas provided at the first substrate. The second device may include a second substrate arranged inside the second device and one second antenna or at least two second antennas provided at the second substrate. In each of the first connection state and the second connection state, when the first display is viewed in a plan view, the first antenna and the second antenna may be arranged at positions superimposed on each other. The first device and the second device may be configured to perform first wireless communication through electromagnetic field coupling between the first antenna and the second antenna. Thus, without providing the external terminal, in each of the first connection state and the second connection state, the first device and the second device can communicate with each other. Therefore, occurrence of high-frequency parasitic emission can be suppressed. Furthermore, necessity for measures against electrostatic discharge can be reduced.

[0007] According to the electronic device according to the above, in each of the first connection state and the second connection state, at least a part of the first antenna may be

arranged at a position superimposed on the second antenna. Strength of electromagnetic field coupling can thus be enhanced.

[0008] According to the electronic device according to the above, one of the first antenna or the second antenna may include a first antenna portion. The other of the first antenna or the second antenna may include a second antenna portion and a third antenna portion. In the first connection state, the first antenna portion may be arranged as being superimposed on the second antenna portion. In the second connection state, the first antenna portion may be arranged as being superimposed on the third antenna portion. The number of antenna portions can thus be set to three. The number of antenna portions can be smaller than in an example where two antenna portions are arranged in the first device and two antenna portions are arranged in the second device.

[0009] According to the electronic device according to the above, the first antenna may include the first antenna portion. The second antenna may include the second antenna portion and the third antenna portion. When the first front surface is viewed in the plan view, the first antenna portion may be arranged at a position adjacent to the first connection portion. When the second rear surface is viewed in the plan view, the second connection portion may be arranged as lying between the second antenna portion and the third antenna portion.

[0010] According to the electronic device according to the above, the first device may include a first substrate arranged inside the first device and one first antenna or at least two first antennas provided at the first substrate. The second device may include a second substrate arranged inside the second device and one second antenna or at least two second antennas provided at the second substrate. One of the first antenna or the second antenna may include a first antenna portion. The other of the first antenna or the second antenna may include a second antenna portion and a third antenna portion. In the first connection state, the second antenna portion may be arranged as being superimposed on the first antenna portion and the third antenna portion may be arranged as not being superimposed on the first antenna. In the second connection state, the third antenna portion may be arranged as being superimposed on the first antenna portion and the second antenna portion may be arranged as not being superimposed on the first antenna.

[0011] According to the electronic device according to the above, when the first front surface is viewed in a plan view, the first antenna portion may be arranged at a position adjacent to the first connection portion. When the second rear surface is viewed in the plan view, the second connection portion may be arranged as lying between the second antenna portion and the third antenna portion.

[0012] According to the electronic device according to the above, in the second connection state, the first device and the second device may be connected with the second display being inclined with respect to the first display.

[0013] According to the electronic device according to the above, in the first connection state, the first device and the second device may be connected with the second display being in parallel to the first display.

[0014] According to the electronic device according to the above, the first connection portion may include a first inclined surface inclined with respect to the first front

surface. The second connection portion may include a second inclined surface inclined with respect to the second rear surface.

[0015] According to the electronic device according to the above, an angle between the first antenna and the second antenna in a cross-section perpendicular to each of the first front surface and the second rear surface in the first connection state is defined as a first angle. An angle between the first antenna and the second antenna in the cross-section perpendicular to each of the first front surface and the second rear surface in the second connection state is defined as a second angle. An angle between the first front surface and the first inclined surface in the cross-section perpendicular to the first front surface is defined as a third angle. An angle between the second rear surface and the second inclined surface in the cross-section perpendicular to the second rear surface is defined as a fourth angle. Each of the first angle and the second angle may be smaller than a total of the third angle and the fourth angle. Thus, in each of the first connection state and the second connection state, high strength of electromagnetic coupling can be maintained.

[0016] According to the electronic device according to the above, when the first antenna and the second antenna are electromagnetically coupled to each other, the first device and the second device may transmit and receive a signal unidirectionally or bidirectionally. The first connection portion may include a first contact for electric power supply. The second connection portion may include a second contact for electric power supply. In each of the first connection state and the second connection state, the first contact and the second contact may be connected. Electric power can thus be supplied with the use of the first contact and the second contact. Since a terminal for electric power supply is less likely to be affected by parasitic emission, wired connection can be established.

[0017] According to the electronic device according to the above, the first device and the second device may be configured to perform second wireless communication while the first device and the second device are separate. Thus, even while the first device and the second device are separate, the first device and the second device can communicate with each other.

[0018] According to the electronic device according to the above, the first wireless communication in the first connection state and the second connection state may be performed through electromagnetic field coupling between the first antenna and the second antenna. The second wireless communication while the first device and the second device are separate may be performed through electromagnetic field coupling between antennas different from each of the first antenna and the second antenna. A frequency band of a signal to be used in the first wireless communication may be higher than a frequency band of a signal to be used in the second wireless communication. In general, as the frequency is higher, higher directivity is demanded. By setting the frequency band of the signal to be used in second wireless communication to be lower than the frequency band of the signal to be used in first wireless communication, communication between the first device and the second device can be ensured even when the first device is distant from the second device.

[0019] According to the electronic device according to the above, the first connection portion may include a first receiving groove and a second receiving groove located

opposite to the first receiving groove. The second connection portion may include a movable hook and a fixed hook located opposite to the movable hook. In the first connection state, the movable hook may be not arranged in the first receiving groove and the fixed hook may be arranged in the second receiving groove. In the second connection state, the movable hook may be arranged in the second receiving groove and the fixed hook may be arranged in the first receiving groove. Thus, in the first connection state, the second device can easily be removed from the first device. In the second connection state, on the other hand, the second device can securely be fixed to the first device.

[0020] According to the electronic device according to the above, the first connection portion may include a first electrical connection portion and a first mechanical connection portion. The first electrical connection portion may be provided above or below the first display. The first mechanical connection portion may be provided to sandwich the first electrical connection portion from a longitudinal direction of the first display. The second connection portion may include a second electrical connection portion and a second mechanical connection portion. The second electrical connection portion may be provided above or below the second display. The second mechanical connection portion may be provided to sandwich the second electrical connection portion from a longitudinal direction of the second display. The first electrical connection portion may electrically be connected to the second electrical connection portion. The first mechanical connection portion may mechanically be connected to the second mechanical connection portion.

[0021] When the mechanical connection portion is provided to sandwich the electrical connection portion from a direction of a short side of the display, an additional space in the direction of the short side is required. On the other hand, there is an extra space at each of opposing sides of the electrical connection portion in the longitudinal direction of the display. As the mechanical connection portion is provided to sandwich the electrical connection portion from the longitudinal direction of the display, the space can effectively be made use of.

[0022] According to the electronic device according to the above, the first electrical connection portion may have terminals disposed in the longitudinal direction of the first display. The second electrical connection portion may have a terminal disposed in the longitudinal direction of the second display. In an example where the terminal of the electrical connection portion is disposed in the longitudinal direction of the display, a length of the display in the direction of the short side can be longer than in an example where the terminal of the electrical connection portion is disposed in the direction of the short side of the display. Consequently, an area of the display can be larger.

[0023] According to the electronic device according to the above, the first device may include an input portion configured to be operated as a stick or a button. The second display may include a touch panel on which a touch input can be provided. Since the first device includes the input portion configured to be operated as the stick or the button, an operation suitable for an application can be performed. In the first connection state, the input portion configured to be operated as the stick or the button is covered with the second device. Therefore, when the electronic device is carried, the input portion is less likely to be caught by something else.

[0024] According to the electronic device according to the above, the first device may include a first housing that surrounds the first display and includes a first projecting portion that extends in an outward direction of the first display. The second device may include a second housing that surrounds the second display and includes a second projecting portion that extends in an outward direction of the second display. The first connection portion may be provided at a front side of the first projecting portion. The second connection portion may be provided at a rear side of the second projecting portion. Thus, the first device and the second device can be connected to each other while the first display and the second display can be increased in size.

[0025] According to the electronic device according to the above, the first device may include a first rear surface, a first upper surface, a first lower surface, a first right side surface, and a first left side surface. The first rear surface may be located opposite to the first front surface. The first upper surface may be contiguous to the first front surface and the first rear surface. The first lower surface may be located opposite to the first upper surface. The first right side surface may be contiguous to each of the first front surface, the first rear surface, the first upper surface, and the first lower surface. The first left side surface may be located opposite to the first right side surface. The second device may include a second upper surface, a second lower surface, a second right side surface, and a second left side surface. The second upper surface may be contiguous to the second front surface and the second rear surface. The second lower surface may be located opposite to the second upper surface. The second right side surface may be contiguous to each of the second front surface, the second rear surface, the second upper surface, and the second lower surface. The second left side surface may be located opposite to the second right side surface. The first projecting portion may be provided at the first upper surface. The second projecting portion may be provided at the second upper surface. The first device may include a first input portion provided at a portion of the first projecting portion at a side of the first right side surface and a second input portion provided at the first upper surface other than the first projecting portion. The second device may include a third input portion provided at a portion of the second projecting portion at a side of the second right side surface and a fourth input portion provided at the second upper surface other than the second projecting portion.

[0026] Thus, a mix-up between the first input portion and the second input portion can be less likely. Similarly, a mix-up between the third input portion and the fourth input portion can be less likely. In the second connection state, the fourth input portion of the second device is arranged at a position distant from the first device. Therefore, interference with operation of the first input portion is less likely. While the first device and the second device are separate, the input portions are provided at positions in common, and hence the first device and the second device can be operated with similar feeling. Furthermore, by horizontally sliding a finger, two different inputs can be provided.

[0027] According to the electronic device according to the above, in the first connection state, in a direction perpendicular to the first front surface, the first input portion and the third input portion may be arranged as being aligned and the second input portion and the fourth input portion may be arranged as being aligned. A user can thus more readily learn positions of the input portions.

[0028] According to the electronic device according to the above, at a surface of the second projecting portion, an input portion or a camera lens may be provided. A functional component that can be used in each of the first connection state and the second connection state can be obtained. Therefore, while the second projecting portion is used as the connection portion, a region in the second device other than the second projecting portion can effectively be made use of. For example, the second display can be increased in size.

[0029] According to the electronic device according to the above, the second display may include a touch panel on which a touch operation can be performed and a touch panel cover on which an input operation to the touch panel can be performed, the touch panel cover covering the touch panel. The second display may include a first tactile feel presentation portion configured with the touch panel cover and a second tactile feel presentation portion where the touch panel is exposed in an opening provided in the touch panel cover. Thus, when the second display includes the touch panel, the user can readily know the position of the second tactile feel presentation portion.

[0030] The foregoing and other objects, features, aspects, and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of an electronic device according to a first embodiment.

[0032] FIG. 2 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of a first device.

[0033] FIG. 3 shows an exemplary illustrative non-limiting drawing of an enlarged schematic diagram of a region III in FIG. 2.

[0034] FIG. 4 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of a second device.

[0035] FIG. 5 shows an exemplary illustrative non-limiting drawing of a partially enlarged view of a second rear surface of the second device.

[0036] FIG. 6 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the electronic device in a first connection state.

[0037] FIG. 7 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line VII-VII in FIG. 6.

[0038] FIG. 8 shows an exemplary illustrative non-limiting drawing of a schematic diagram showing relation between a first connection portion and a second connection portion in the first connection state.

[0039] FIG. 9 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line IX-IX in FIG. 6.

[0040] FIG. 10 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the electronic device in a second connection state.

[0041] FIG. 11 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XI-XI in FIG. 10.

[0042] FIG. 12 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a state in which the electronic device in the second connection state is unlocked.

[0043] FIG. 13 shows an exemplary illustrative non-limiting drawing of a schematic diagram showing relation between the first connection portion and the second connection portion in the second connection state.

[0044] FIG. 14 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing positional relation between a first antenna and a second antenna in the first connection state.

[0045] FIG. 15 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing positional relation between the first antenna and the second antenna in the second connection state.

[0046] FIG. 16 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a modification of positional relation between the first antenna and the second antenna in the first connection state.

[0047] FIG. 17 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a modification of positional relation between the first antenna and the second antenna in the second connection state.

[0048] FIG. 18 shows an exemplary illustrative non-limiting drawing of a partial schematic perspective view showing a configuration of the electronic device according to a second embodiment.

[0049] FIG. 19 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XIX-XIX in FIG. 18.

[0050] FIG. 20 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the first device of the electronic device according to a third embodiment.

[0051] FIG. 21 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a configuration of a fifth input portion and a sixth input portion.

[0052] FIG. 22 shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the second device of the electronic device according to the third embodiment.

[0053] FIG. 23 shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XXIII-XXIII in FIG. 22.

[0054] FIG. 24 shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of the electronic device according to the third embodiment in the first connection state.

[0055] FIG. 25 shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of the electronic device according to the third embodiment in the second connection state.

DETAILED DESCRIPTION OF NON-LIMITING EXAMPLE EMBODIMENTS

[0056] An embodiment of the present disclosure will be described in detail with reference to the drawings. The same or corresponding elements in the drawings have the same reference characters allotted and description thereof will not be repeated.

First Embodiment

[0057] A configuration of an electronic device 300 according to a first embodiment will initially be described.

[0058] FIG. 1 is a schematic perspective view showing the configuration of electronic device 300 according to the first embodiment. As shown in FIG. 1, electronic device 300 according to the first embodiment includes a first device 100 and a second device 200. First device 100 and second device 200 are removably attachable to each other. Second device 200 is attachable to first device 100 and removable from first device 100.

[0059] As shown in FIG. 1, first device 100 includes a first front surface 101 and a first rear surface 102. First rear surface 102 is located opposite to first front surface 101. Second device 200 includes a second front surface 201, a second rear surface 202, and a second display 210. Second rear surface 202 is located opposite to second front surface 201. Second display 210 is arranged at second front surface 201. Second display 210 may be a touch panel on which a touch input can be provided.

[0060] FIG. 2 is a schematic front view showing a configuration of first device 100. First device 100 includes a first upper surface 103, a first lower surface 104, a first right side surface 105, and a first left side surface 106. First upper surface 103 is contiguous to each of first front surface 101, first rear surface 102, first left side surface 106, and first right side surface 105. First lower surface 104 is located opposite to first upper surface 103. First right side surface 105 is contiguous to each of first front surface 101, first rear surface 102, first upper surface 103, and first left side surface 106. First left side surface 106 is located opposite to first right side surface 105. First left side surface 106 is contiguous to each of first front surface 101, first rear surface 102, first upper surface 103, and first lower surface 104.

[0061] As shown in FIG. 2, first device 100 may further include a first housing 165, a first display 110, a first connection portion 120, a ninth input portion 121, an operation button 190, an unlocking button 191, and a Hall element 130. First display 110 is arranged at first front surface 101. First display 110 is attached to first housing 165.

[0062] First connection portion 120 is arranged at first front surface 101. First connection portion 120 is attached to first housing 165. First connection portion 120 may project from a surface of first housing 165. When first display 110 is viewed in a plan view, first connection portion 120 is arranged between first upper surface 103 and first display 110. First display 110 being viewed in the plan view corresponds to first device 100 being viewed in a direction from first front surface 101 toward first rear surface 102.

[0063] As shown in FIG. 2, ninth input portion 121 is arranged, for example, at first front surface 101. Ninth input portion 121 can be operated as a stick or a button. Unlocking button 191 is arranged, for example, at first upper surface 103. Operation button 190 is arranged, for example, at first upper surface 103. Hall element 130 is arranged inside first housing 165. First device 100 does not have to include ninth input portion 121.

[0064] FIG. 3 is an enlarged schematic diagram of a region III in FIG. 2. As shown in FIG. 3, first connection portion 120 includes, for example, a first mechanical connection portion 180, a first electrical connection portion 160, and a first magnetic connection portion 111. First mechanical connection portion 180 includes a first main body portion 183 and a pair of first protruding portions 184. First main

body portion 183 is attached to first housing 165. The pair of first protruding portions 184 is provided at first main body portion 183. First main body portion 183 is provided with a first opening portion 185.

[0065] First electrical connection portion 160 is exposed through first opening portion 185. First electrical connection portion 160 includes a first contact 162 and a first holding portion 161. First contact 162 is conductive. First holding portion 161 is insulating. First contact 162 is held by first holding portion 161. Though the number of first contacts 162 is not particularly limited, for example, nine first contacts are provided. When first display 110 is viewed in the plan view, first contacts 162 are arranged along a line. One of first contacts 162 is, for example, a terminal for electric power supply.

[0066] First magnetic connection portion 111 is configured, for example, with a pair of magnets. First magnetic connection portion 111 is located between first main body portion 183 and first rear surface 102. First magnetic connection portion 111 is arranged inside first device 100. As shown in FIG. 3, when first front surface 101 is viewed in the plan view, first electrical connection portion 160 is located as lying between the pair of magnets. In the present embodiment, when first front surface 101 is viewed in the plan view, first electrical connection portion 160 is located as lying between the pair of magnets in a longitudinal direction of first display 110 of first device 100.

[0067] First device 100 includes a first substrate 131 and a first antenna 132. Each of first substrate 131 and first antenna 132 is arranged inside first device 100. More particularly, each of first substrate 131 and first antenna 132 is arranged inside first housing 165. First antenna 132 is provided at first substrate 131.

[0068] The number of first antennas 132 is not particularly limited. The number of first antennas 132 is set, for example, to one or at least two. In electronic device 300 according to the present embodiment, a single first antenna 132 is provided. First antenna 132 is, for example, a first antenna portion 132. When first front surface 101 is viewed in the plan view, first antenna portion 132 may be arranged at a position adjacent to first connection portion 120. First antenna portion 132 is arranged, for example, between first connection portion 120 and first right side surface 105.

[0069] As shown in FIG. 3, when first front surface 101 is viewed in the plan view, first antenna 132 may be located on an outer side of first connection portion 120. From another point of view, when first front surface 101 is viewed in the plan view, first antenna 132 may be arranged at a position not superimposed on first main body portion 183. For example, resin is employed as a material for first housing 165. Though the material for first main body portion 183 is not particularly limited, for example, metal may be adopted. First antenna 132 may be arranged such that first main body portion 183 does not cut off electromagnetic field.

[0070] FIG. 4 is a schematic front view showing a configuration of second device 200. As shown in FIG. 4, second device 200 further includes a second upper surface 203, a second lower surface 204, a second right side surface 205, and a second left side surface 206. Second upper surface 203 is contiguous to second front surface 201 and second rear surface 202. Second lower surface 204 is located opposite to second upper surface 203. Second right side surface 205 is contiguous to each of second front surface 201, second rear surface 202, second upper surface 203, and second lower

surface 204. Second left side surface 206 is located opposite to second right side surface 205. Second left side surface 206 is contiguous to each of second front surface 201, second rear surface 202, second upper surface 203, and second lower surface 204.

[0071] As shown in FIG. 4, second device 200 includes a second housing 265 and a magnet portion 230. Second display 210 is attached to second housing 265. Magnet portion 230 is arranged inside second housing 265. As shown in FIG. 4, when second display 210 is viewed in the plan view, magnet portion 230 may be located, for example, between second display 210 and second right side surface 205. Second display 210 being viewed in the plan view corresponds to second device 200 being viewed in a direction from second front surface 201 toward second rear surface 202.

[0072] FIG. 5 is a partially enlarged view of second rear surface 202 of second device 200. As shown in FIG. 5, second device 200 includes a second connection portion 220. Second connection portion 220 is arranged at second rear surface 202. Second connection portion 220 includes, for example, a second mechanical connection portion 280, a second electrical connection portion 260, and a second magnetic connection portion 211. Second mechanical connection portion 280 includes a second main body portion 283, a fixed hook 281, and a movable hook 270. Second main body portion 283 is attached to second housing 265. Fixed hook 281 is fixed to second main body portion 283. Movable hook 270 is attached to second main body portion 283 as being movable.

[0073] Second main body portion 283 is provided with a second opening portion 285 and a third opening portion 284. Movable hook 270 is exposed through third opening portion 284. Movable hook 270 is located opposite to fixed hook 281. When second rear surface 202 is viewed in the plan view, second electrical connection portion 260 is located between movable hook 270 and fixed hook 281. Second electrical connection portion 260 is exposed through second opening portion 285. When second rear surface 202 is viewed in the plan view, fixed hook 281 is located between second upper surface 203 and movable hook 270.

[0074] Second electrical connection portion 260 includes a second contact 262 and a second holding portion 261. Second contact 262 is conductive. Second holding portion 261 is insulating. Second contact 262 is held by second holding portion 261. Though the number of second contacts 262 is not particularly limited, for example, nine second contacts are provided. When second rear surface 202 is viewed in the plan view, second contacts 262 are arranged along a line. One of second contacts 262 is, for example, a terminal for electric power supply.

[0075] Second magnetic connection portion 211 is configured, for example, with a pair of magnets. Second magnetic connection portion 211 is located between second main body portion 283 and second front surface 201. Second magnetic connection portion 211 is arranged inside second device 200. When second rear surface 202 is viewed in the plan view, second electrical connection portion 260 is located as lying between the pair of magnets. Second magnetic connection portion 211 is magnetically attractable to first magnetic connection portion 111 while it is distant from first magnetic connection portion 111.

[0076] As shown in FIG. 5, second device 200 includes a second substrate 231 and a second antenna 232. Each of

second substrate 231 and second antenna 232 is arranged inside second device 200. More particularly, each of second substrate 231 and second antenna 232 is arranged inside second housing 265. Second antenna 232 is provided at second substrate 231.

[0077] The number of second antennas 232 is not particularly limited. The number of second antennas 232 is set, for example, to one or at least two. In electronic device 300 according to the present embodiment, two second antennas 232 are provided. Second antenna 232 may include a second antenna portion 234 and a third antenna portion 236. Second substrate 231 may include a first substrate portion 233 and a second substrate portion 235. Second antenna portion 234 is provided at first substrate portion 233. Third antenna portion 236 is provided at second substrate portion 235. When second rear surface 202 is viewed in the plan view, second connection portion 220 may be arranged as lying between second antenna portion 234 and third antenna portion 236.

[0078] As shown in FIG. 5, when second rear surface 202 is viewed in the plan view, second antenna 232 may be located on an outer side of second connection portion 220 in each of directions toward left and right side surfaces. From another point of view, when second rear surface 202 is viewed in the plan view, second antenna 232 may be arranged at a position not superimposed on second main body portion 283. For example, resin is employed as a material for second housing 265. Though the material for second main body portion 283 is not particularly limited, for example, metal may be adopted. Second antenna 232 may be arranged such that second main body portion 283 does not cut off electromagnetic field.

[0079] FIG. 6 is a schematic front view showing a configuration of electronic device 300 in a first connection state. In the first connection state, second rear surface 202 is arranged to cover first display 110. Second rear surface 202 may be arranged to cover the entire surface or at least a part of first display 110. Second rear surface 202 is arranged as being superimposed on first front surface 101. Second front surface 201 faces outward. A user can visually recognize second display 210 arranged at second front surface 201. In the first connection state, Hall element 130 is arranged as being superimposed on magnet portion 230. With Hall element 130, the current connection state may be determined as the first connection state.

[0080] A direction from first left side surface 106 toward first right side surface 105 of first device 100 corresponds to a reference direction B. In the first connection state, when first display 110 is viewed in the plan view, a direction from second left side surface 206 toward second right side surface 205 of second device 200 is a first direction A1. In the first connection state, first device 100 and second device 200 are connected such that the direction from second left side surface 206 toward second right side surface 205 of second device 200 is the same as reference direction B when first display 110 is viewed in the plan view.

[0081] FIG. 7 is a schematic cross-sectional view along the line VII-VII in FIG. 6. First connection portion 120 is connected to second connection portion 220. Specifically, first mechanical connection portion 180 is mechanically connected to second mechanical connection portion 280. First mechanical connection portion 180 includes a first receiving groove 181 and a second receiving groove 182. Specifically, each of first receiving groove 181 and second

receiving groove 182 is provided at a rear side of first main body portion 183. Second receiving groove 182 is located opposite to first receiving groove 181. First electrical connection portion 160 is located between first receiving groove 181 and second receiving groove 182.

[0082] First connection portion 120 is provided with a fourth opening portion 186. Unlocking button 191 is exposed through fourth opening portion 186. First device 100 includes a first rotation shaft 192. Unlocking button 191 is pivotable around first rotation shaft 192. Fixed hook 281 is arranged in second receiving groove 182 provided at the rear side of first main body portion 183. Unlocking button 191 may be in contact with fixed hook 281.

[0083] Second device 200 includes a second rotation shaft 273 and a biasing member 274. Movable hook 270 is pivotable around second rotation shaft 273. Movable hook 270 includes a first tab portion 271 and a first support portion 272. First tab portion 271 is contiguous to first support portion 272. First support portion 272 is pivotably attached to second rotation shaft 273. Biasing member 274 biases movable hook 270. Biasing member 274 is, for example, a spring. Each of second rotation shaft 273 and biasing member 274 is arranged inside second device 200. A part of movable hook 270 passes through third opening portion 284.

[0084] As shown in FIGS. 3 and 5, a width (a second width W2) of movable hook 270 is larger than a width (a first width W1) of a pair of first protruding portions 184. Therefore, in the first connection state, movable hook 270 rides on the pair of first protruding portions 184. In other words, movable hook 270 is not arranged in first receiving groove 181. As shown in FIG. 7, in the first connection state, as fixed hook 281 enters second receiving groove 182, first device 100 and second device 200 are connected. Second device 200 is thus attached to first device 100. In removal of second device 200 from first device 100, second device 200 is turned in a first rotation direction R1. In this case, unlocking button 191 does not have to be pushed.

[0085] As shown in FIG. 7, electronic device 300 includes a fourth protruding portion 9. Fourth protruding portion 9 may be attached to first front surface 101 of first device 100 or second rear surface 202 of second device 200. In the first connection state, fourth protruding portion 9 is located between first front surface 101 and second rear surface 202. In the first connection state, first front surface 101 and second rear surface 202 are substantially in parallel to each other. In the first connection state, while second display 210 is in parallel to first display 110, first device 100 and second device 200 may be connected. With fourth protruding portion 9, a gap is provided between first front surface 101 and second rear surface 202. Ninth input portion 121 is distant from second rear surface 202. Fourth protruding portion 9 can prevent first device 100 and second device 200 from wobbling in the first connection state.

[0086] As shown in FIG. 7, first electrical connection portion 160 is electrically connected to second electrical connection portion 260. FIG. 8 is a schematic diagram showing relation between the first connection portion and the second connection portion in the first connection state. The first connection state refers to a state in which second connection portion 220 is connected to first connection portion 120 in first direction A1. In the first connection state, first contact 162 is electrically connected to second contact 262.

[0087] As shown in FIG. 8, first contact 162 includes a first terminal 162a, a second terminal 162b, and a third terminal 162c. First electrical connection portion 160 may have terminals disposed in the longitudinal direction of first display 110. First terminal 162a is located closest to the first left side surface. Third terminal 162c is located closest to the first right side surface. Second terminal 162b is located intermediate between first terminal 162a and third terminal 162c. A direction from first terminal 162a toward third terminal 162c corresponds to reference direction B.

[0088] Second contact 262 includes a fourth terminal 262a, a fifth terminal 262b, and a sixth terminal 262c. Second electrical connection portion 260 may have terminals disposed in the longitudinal direction of second display 210. Fourth terminal 262a is located closest to the second left side surface. Sixth terminal 262c is located closest to the second right side surface. Fifth terminal 262b is located intermediate between fourth terminal 262a and sixth terminal 262c. In the first connection state, first terminal 162a, second terminal 162b, and third terminal 162c are electrically connected to fourth terminal 262a, fifth terminal 262b, and sixth terminal 262c, respectively. In the first connection state, second connection portion 220 is arranged with respect to first connection portion 120 such that a direction from fourth terminal 262a toward sixth terminal 262c is the same as first direction A1. Without being limited to such arrangement of the terminals, first contact 162 and second contact 262 should only be in arrangement relation in which they are connected to each other in the first connection state and the second connection state. For example, a plurality of terminals may be arranged such that, with respect to a terminal arranged at the center, other terminals are in line symmetry.

[0089] FIG. 9 is a schematic cross-sectional view along the line IX-IX in FIG. 6. As shown in FIG. 9, first magnetic connection portion 111 is arranged inside first main body portion 183. First main body portion 183 includes a first inclined surface 187. First inclined surface 187 is inclined with respect to first front surface 101 of first housing 165. An angle of inclination of first inclined surface 187 with respect to first front surface 101 is, for example, 10°. First magnetic connection portion 111 is arranged along first inclined surface 187. A surface of first magnetic connection portion 111 may substantially be in parallel to first inclined surface 187.

[0090] Similarly, second magnetic connection portion 211 is arranged inside second main body portion 283. Second main body portion 283 includes a second inclined surface 287. Second inclined surface 287 is inclined with respect to second rear surface 202 of second housing 265. An angle of inclination of second inclined surface 287 with respect to second rear surface 202 is, for example, 10°. Second magnetic connection portion 211 is arranged along second inclined surface 287. A surface of second magnetic connection portion 211 may substantially be in parallel to second inclined surface 287. The surface of second magnetic connection portion 211 may substantially be in parallel to the surface of first magnetic connection portion 111.

[0091] FIG. 10 is a schematic front view showing a configuration of electronic device 300 in the second connection state. In the second connection state, second rear surface 202 is arranged not to cover first display 110. Therefore, the user can visually recognize each of first display 110 and second display 210. In the second connection

state, a part of first front surface 101 of first device 100 is arranged as being superimposed on a part of second rear surface 202 of second device 200. In the second connection state, Hall element 130 is arranged as not being superimposed on magnet portion 230. With Hall element 130, the current connection state may be determined as the second connection state.

[0092] In the second connection state, when first display 110 is viewed in the plan view, the direction from second left side surface 206 toward second right side surface 205 of second device 200 is a second direction A2. In the second connection state, first device 100 and second device 200 are connected such that the direction from second left side surface 206 toward second right side surface 205 of second device 200 is opposite to reference direction B when first display 110 is viewed in the plan view.

[0093] FIG. 11 is a schematic cross-sectional view along the line XI-XI in FIG. 10. As shown in FIG. 11, in the second connection state, movable hook 270 is arranged in second receiving groove 182 and fixed hook 281 is arranged in first receiving groove 181. Movable hook 270 has entered second receiving groove 182. Fixed hook 281 has entered first receiving groove 181. First mechanical connection portion 180 is thus mechanically fixed to second mechanical connection portion 280.

[0094] As shown in FIG. 11, first electrical connection portion 160 is electrically connected to second electrical connection portion 260. Specifically, in the second connection state, first contact 162 and second contact 262 are connected. Second rear surface 202 is inclined with respect to first front surface 101. In the second connection state, first device 100 and second device 200 may be connected while second display 210 is inclined with respect to first display 110. An angle of inclination of second rear surface 202 with respect to first front surface 101 is, for example, 20°.

[0095] FIG. 12 is a schematic cross-sectional view showing a state in which electronic device 300 in the second connection state is unlocked. As shown in FIG. 12, unlocking button 191 is pushed into first device 100 along a third direction A3. As unlocking button 191 pushes movable hook 270, movable hook 270 is pushed out of second receiving groove 182. Consequently, mechanical locking is released. When second device 200 is turned in second rotation direction R2 in this state, second device 200 is separated from first device 100.

[0096] FIG. 13 is a schematic diagram showing relation between the first connection portion and the second connection portion in the second connection state. The second connection state refers to a state in which second connection portion 220 is connected to first connection portion 120 in second direction A2. Second direction A2 is opposite to first direction A1. In the second connection state, first terminal 162a, second terminal 162b, and third terminal 162c are electrically connected to sixth terminal 262c, fifth terminal 262b, and fourth terminal 262a, respectively. In the second connection state, second connection portion 220 is arranged with respect to first connection portion 120 such that the direction from fourth terminal 262a toward sixth terminal 262c is the same as second direction A2.

[0097] Each of second terminal 162b and fifth terminal 262b is, for example, a terminal for electric power supply. In each of the first connection state and the second connection state, second terminal 162b is electrically connected to fifth terminal 262b. Usage of first terminal 162a may be the same

as usage of third terminal 162c. Similarly, usage of fourth terminal 262a may be the same as usage of sixth terminal 262c. In this case, the same function can be performed in the first connection state and the second connection state. In the first connection state and the second connection state, first contact 162 and second contact 262 may define a plurality of corresponding electrical communication paths. Though usage of first terminal 162a, third terminal 162c, fourth terminal 262a, and sixth terminal 262c is not particularly limited, those terminals may be, for example, terminals for grounding, terminals for determination as to whether or not first device 100 and second device 200 are connected, or terminals for resetting the display.

[0098] FIG. 14 is a schematic cross-sectional view showing positional relation between first antenna 132 and second antenna 232 in the first connection state. As shown in FIG. 14, in the first connection state, first antenna 132 is arranged as being superimposed on second antenna portion 234 of second antenna 232. In other words, in the first connection state, when first display 110 is viewed in the plan view, first antenna 132 and second antenna 232 are arranged at positions superimposed on each other. Each of first antenna 132 and second antenna 232 is a module in a form of a plate.

[0099] As shown in FIG. 14, in a cross-section perpendicular to first front surface 101, first antenna 132 is substantially in parallel to first front surface 101. First antenna 132 is inclined with respect to first inclined surface 187. In a cross-section perpendicular to second rear surface 202, second antenna portion 234 of second antenna 232 is substantially in parallel to second rear surface 202. Second antenna portion 234 of second antenna 232 is inclined with respect to second inclined surface 287. In the first connection state, second antenna portion 234 of second antenna 232 is substantially in parallel to first antenna 132.

[0100] In the first connection state, second inclined surface 287 is in contact with first inclined surface 187. An angle between first front surface 101 and first inclined surface 187 in the cross-section perpendicular to first front surface 101 is defined as a third angle $\theta 3$. Third angle $\theta 3$ is, for example, 10° . An angle between second rear surface 202 and second inclined surface 287 in the cross-section perpendicular to second rear surface 202 is defined as a fourth angle $\theta 4$. Fourth angle $\theta 4$ is, for example, 10° .

[0101] FIG. 15 is a schematic cross-sectional view showing positional relation between first antenna 132 and second antenna 232 in the second connection state. As shown in FIG. 15, in the second connection state, first antenna portion 132 of first antenna 132 is arranged as being superimposed on third antenna portion 236 of second antenna 232. In other words, in the second connection state, when first display 110 is viewed in the plan view, first antenna 132 and second antenna 232 are arranged at positions superimposed on each other. As set forth above, in each of the first connection state and the second connection state, at least a part of first antenna 132 is arranged at a position superimposed on second antenna 232.

[0102] As shown in FIG. 15, in the second connection state, third antenna portion 236 of second antenna 232 is inclined with respect to first antenna 132. In the cross-section perpendicular to each of first front surface 101 and second rear surface 202, an angle of inclination (a second angle $\theta 2$) of third antenna portion 236 of second antenna 232 with respect to first antenna 132 is, for example, 20° . In the

second connection state, second inclined surface 287 is in contact with first inclined surface 187.

[0103] Though an example in which first inclined surface 187 and second inclined surface 287 are provided is described with reference to FIGS. 14 and 15, first inclined surface 187 does not have to be provided in first mechanical connection portion 180 or second inclined surface 287 does not have to be provided in second mechanical connection portion 280. Specifically, first mechanical connection portion 180 and second mechanical connection portion 280 may be configured to be connected in parallel to each other. In that case, first device 100 and second device 200 may be in parallel to each other, or one of first device 100 or second device 200 may be inclined with respect to the other of first device 100 or second device 200.

[0104] Though the configuration in which first antenna 132 includes a single antenna portion and second antenna 232 includes two antenna portions is described above, the present disclosure is not limited to the configuration. In electronic device 300 according to another manner, first antenna 132 may include two antenna portions and second antenna 232 may include a single antenna portion. Specifically, second antenna 232 may include the first antenna portion. First antenna 132 may include the second antenna portion and the third antenna portion. In the first connection state, the first antenna portion may be arranged as being superimposed on the second antenna portion. In the second connection state, the first antenna portion may be arranged as being superimposed on the third antenna portion. In electronic device 300 according to yet another manner, first antenna 132 may include two antenna portions and second antenna 232 may include two antenna portions.

[0105] In each of the first connection state and the second connection state, first device 100 and second device 200 can perform first wireless communication through electromagnetic field coupling between first antenna 132 and second antenna 232. When first antenna 132 and second antenna 232 are electromagnetically coupled to each other, first device 100 and second device 200 may bidirectionally transmit and receive a signal. Second device 200 may unidirectionally receive a signal transmitted from first device 100. First device 100 may unidirectionally receive a signal transmitted from second device 200. Though each of first antenna 132 and second antenna 232 is not particularly limited, it is, for example, a 60-GHz millimeter wave module.

Modification of First Embodiment

[0106] FIG. 16 is a schematic cross-sectional view showing a modification of positional

[0107] relation between first antenna 132 and second antenna 232 in the first connection state. As shown in FIG. 16, in the cross-section perpendicular to first front surface 101, first antenna 132 may be inclined with respect to first front surface 101. In the cross-section perpendicular to first front surface 101, first antenna 132 may be inclined with respect to first inclined surface 187. Second antenna portion 234 of second antenna 232 may substantially be in parallel to second rear surface 202. In the first connection state, an angle between first antenna 132 and second antenna 232 in the cross-section perpendicular to each of first front surface 101 and second rear surface 202 is defined as a first angle $\theta 1$. First angle $\theta 1$ may be smaller than a total of third angle $\theta 3$

and fourth angle θ_4 . Though first angle θ_1 is not particularly limited, it is, for example, not smaller than 2° and not larger than 18° .

[0108] FIG. 17 is a schematic cross-sectional view showing a modification of positional relation between first antenna 132 and second antenna 232 in the second connection state. As shown in FIG. 17, in the second connection state, an angle between first antenna 132 and third antenna portion 236 of second antenna 232 in the cross-section perpendicular to each of first front surface 101 and second rear surface 202 is defined as second angle θ_2 . Second angle θ_2 may be smaller than the total of third angle θ_3 and fourth angle θ_4 . Though second angle θ_2 is not particularly limited, it may be smaller than the total of third angle θ_3 and fourth angle θ_4 . For example, second angle θ_2 may be half the total of third angle θ_3 and fourth angle θ_4 . In the present example, second angle θ_2 is 10° . Second angle θ_2 may be not smaller than 2° and not larger than 18° . According to such a configuration, in the first connection state and the second connection state, first antenna 132 and second antenna 232 can be closer to being in parallel. Therefore, for example, in a high frequency band where directivity is strong, a signal can be received with good sensitivity.

[0109] First angle θ_1 may be substantially the same as or different from second angle θ_2 . Though an absolute value of a difference between first angle θ_1 and second angle θ_2 is not particularly limited, it may be not larger than 15° or not larger than 10° .

[0110] Though an example in which first antenna 132 is inclined with respect to first front surface 101 and second antenna 232 is substantially in parallel to second rear surface 202 is described above, electronic device 300 in the present disclosure is not limited as such. Specifically, first antenna 132 may substantially be in parallel to first front surface 101 and second antenna 232 may be inclined with respect to second rear surface 202, or first antenna 132 may be inclined with respect to first front surface 101 and second antenna 232 may be inclined with respect to second rear surface 202.

Method of Using Electronic Device

[0111] A method of using electronic device 300 will now be described. Electronic device 300 is, for example, a game console, a smartphone, a tablet personal computer, or the like. First device 100 and second device 200 are usable in both of a case where they are connected to each other and a case where they are separate from each other. As shown in FIGS. 6 and 7, in the first connection state, first device 100 and second device 200 are connectable. As shown in FIG. 8, in the first connection state, second connection portion 220 is oriented in first direction A1 with respect to first connection portion 120. As shown in FIGS. 10 and 11, in the second connection state, first device 100 and second device 200 are connectable. As shown in FIG. 13, in the second connection state, second connection portion 220 is oriented in second direction A2 with respect to first connection portion 120. In the first connection state and the second connection state, first wireless communication is performed through electromagnetic field coupling between first antenna 132 and second antenna 232.

[0112] In the first connection state and the second connection state, first device 100 may transmit an image signal to second device 200 through first antenna 132 and second antenna 232. The image signal is, for example, a scalable low voltage signaling (SLVS) signal. For example, in first

device 100, a mobile industry processor interface (MIPI)-display serial interface (DSI) signal is converted to the SLVS signal. The converted SLVS signal is sent from first device 100 to second device 200. In second device 200, the SLVS signal is converted to the MIPI-DSI signal. In another manner, the MIPI-DSI signal may directly be transmitted from first device 100 to second device 200 without being converted to the SLVS signal.

[0113] While first device 100 and second device 200 are separate, first device 100 and second device 200 may be able to perform second wireless communication. Second wireless communication while first device 100 and second device 200 are separate may be performed through electromagnetic field coupling between antennas different from each of first antenna 132 and second antenna 232. Specifically, second wireless communication may be, for example, Wifi® communication. A frequency band of a signal to be used in first wireless communication may be higher than a frequency band of a signal to be used in second wireless communication.

[0114] Second wireless communication is not limited to Wifi® communication. Second wireless communication may be, for example, communication based on Bluetooth® or infrared communication. Second device 200 may receive a signal from first device 100 and show an image, or second device 200 itself may generate an image without receiving a signal from first device 100.

[0115] In the first connection state, the user can play a game in a single-screen mode through first wireless communication. In the second connection state, the user can play a game in a dual-screen mode through first wireless communication. While first device 100 and second device 200 are separate, a first user uses first device 100 and a second user uses second device 200. The first user and the second user can simultaneously play the same game through second wireless communication. In another method of use, one user can operate first device 100 while viewing a screen of second device 200, with the second device being at rest.

[0116] At least one of first device 100 and second device 200 that configure electronic device 300 may further include a storage (not shown) and/or a processor (not shown). The storage is, for example, a dynamic random access memory (DRAM) or a non-volatile memory. An application program such as a game may be stored in the storage. The processor may be able to perform information processing by reading the application program. First display 110 and second display 210 may show an image, for example, generated as a result of information processing performed by the processor. Electronic device 300 may be an information processing apparatus other than the game console.

Second Embodiment

[0117] A configuration of electronic device 300 according to a second embodiment will now be described. Electronic device 300 according to the second embodiment is different from electronic device 300 according to the first embodiment mainly in configuration of first mechanical connection portion 180 and second mechanical connection portion 280, and it is otherwise substantially the same as electronic device 300 according to the first embodiment. A configuration different from that of electronic device 300 according to the first embodiment will mainly be described below.

[0118] FIG. 18 is a partial schematic perspective view showing the configuration of electronic device 300 accord-

ing to the second embodiment. FIG. 19 is a schematic cross-sectional view along the line XIX-XIX in FIG. 18. As shown in FIGS. 18 and 19, first mechanical connection portion 180 includes a movable member 140, a fixed member 150, and a pair of third biasing portions 154. Movable member 140 includes a first plate member 141 and a pair of second plate members 142. When first front surface 101 is viewed in the plan view, first plate member 141 is in a trapezoidal shape. The pair of second plate members 142 is arranged at opposing sides of first plate member 141. The pair of third biasing portions 154 biases the pair of second plate members 142 in a direction toward first plate member 141, respectively.

[0119] Fixed member 150 includes an upper surface plate 151 and a partition plate 152. Upper surface plate 151 is arranged above first magnetic connection portion 111. Partition plate 152 is attached to upper surface plate 151. Partition plate 152 extends from upper surface plate 151 toward first rear surface 102. First magnetic connection portion 111 is arranged between the pair of partition plates 152.

[0120] Second mechanical connection portion 280 includes a pair of movable hook portions 250, a pair of rotation shaft portions 253, a pair of fourth biasing portions 254, and a lower surface plate 255. The pair of movable hook portions 250 can pivot around the pair of rotation shaft portions 253, respectively. Each of the pair of movable hook portions 250 includes a second tab portion 251 and a second support portion 252. Second tab portion 251 is contiguous to second support portion 252. Second support portion 252 is pivotably attached to rotation shaft portion 253. The pair of fourth biasing portions 254 biases the pair of movable hook portions 250, respectively. Lower surface plate 255 is arranged below second magnetic connection portion 211.

[0121] As shown in FIG. 19, when unlocking button 191 is pushed into first device 100 along third direction A3, first plate member 141 pushes the pair of second plate members 142 toward opposing sides. Each of the pair of second plate members 142 moves in a fourth direction A4 perpendicular to third direction A3. The pair of second plate members 142 thus pushes second tab portions 251 of the pair of movable hook portions 250 outward. Consequently, the pair of movable hook portions 250 is unlocked. The pair of unlocked movable hook portions 250 is shown with a dashed line.

[0122] As the user releases unlocking button 191, each of the pair of second plate members 142 moves to its original position owing to third biasing portion 154. Each of the pair of movable hook portions 250 moves to its original position owing to fourth biasing portion 254. Second tab portion 251 of each of the pair of movable hook portions 250 moves to a lower side of upper surface plate 151. Consequently, second mechanical connection portion 280 is fixed to first mechanical connection portion 180. The pair of locked movable hook portions 250 is shown with a solid line.

[0123] First electrical connection portion 160 may be provided above or below first display 110. First mechanical connection portion 180 is provided to sandwich first electrical connection portion 160 from the longitudinal direction of first display 110. From another point of view, first mechanical connection portion 180 is provided at opposing sides of first electrical connection portion 160.

[0124] Second electrical connection portion 260 may be provided above or below second display 210. Second mechanical connection portion 280 is provided to sandwich

second electrical connection portion 260 from the longitudinal direction of second display 210. From another point of view, second mechanical connection portion 280 is provided at opposing sides of second electrical connection portion 260. First electrical connection portion 160 is electrically connected to second electrical connection portion 260. First mechanical connection portion 180 is mechanically connected to second mechanical connection portion 280.

[0125] In the above, unlocking button 191 is pushed in third direction A3 so that each of the pair of second plate members 142 slides in fourth direction A4 and each of the pair of movable hook portions 250 is unlocked. A locking mechanism is common to the first connection state and the second connection state. Though the configuration in which a slide mechanism releases locking is described above, an unlocking mechanism in the present disclosure is not limited to the slide mechanism. A link mechanism or a gear mechanism may be employed instead of the slide mechanism.

Third Embodiment

[0126] A configuration of electronic device 300 according to a third embodiment will now be described. Electronic device 300 according to the third embodiment is different from electronic devices 300 according to the first embodiment and the second embodiment mainly in including a first projecting portion 10 and a second projecting portion 20, and it is otherwise substantially the same as electronic devices 300 according to the first embodiment and the second embodiment. A configuration different from that of electronic devices 300 according to the first embodiment and the second embodiment will mainly be described below.

[0127] FIG. 20 is a schematic front view showing a configuration of first device 100 of electronic device 300 according to the third embodiment. As shown in FIG. 20, when first display 110 is viewed in the plan view, first housing 165 surrounds first display 110. First housing 165 includes first projecting portion 10. First projecting portion 10 extends in an outward direction of first display 110. First connection portion 120 is provided at a front side of first projecting portion 10.

[0128] First device 100 of electronic device 300 shown in FIG. 20 is, for example, a game console, a smartphone, a tablet personal computer, a controller that can transmit an input signal to another electronic device, or the like. First device 100 of electronic device 300 does not have to include first display 110.

[0129] First upper surface 103 includes a part of a sixth input portion 6, a first left upper surface portion 17, first projecting portion 10, a first right upper surface portion 16, and a part of a second input portion 2. First projecting portion 10 is formed by connection of a first right inclined portion 12, a first left inclined portion 13, a first central portion 11, a first front surface portion 14 which is a part of first front surface 101, and a first rear surface portion which is a part of first rear surface 102. First projecting portion 10 is distant from each of first right side surface 105, first left side surface 106, and first lower surface 104. First projecting portion 10 may project only from a portion in first upper surface 103 around a portion intermediate between first front surface 101 and first rear surface 102 and may be distant from each of first front surface 101 and first rear surface 102.

[0130] First projecting portion 10 includes first right inclined portion 12, first left inclined portion 13, first central portion 11, and first front surface portion 14. First right

inclined portion 12 is located at a first right side surface 105 side of first projecting portion 10. First left inclined portion 13 is located at a first left side surface 106 side of first projecting portion 10. First central portion 11 is located between first right inclined portion 12 and first left inclined portion 13. First central portion 11 is contiguous to each of first right inclined portion 12 and first left inclined portion 13. First front surface portion 14 is contiguous to each of first right inclined portion 12, first left inclined portion 13, and first central portion 11. First front surface portion 14 defines a part of first front surface 101.

[0131] As shown in FIG. 20, when first display 110 is viewed in the plan view, first central portion 11 is substantially in parallel to first lower surface 104. Each of first right inclined portion 12 and first left inclined portion 13 is inclined with respect to first central portion 11. A boundary portion between first central portion 11 and first right inclined portion 12 may be curved as convexly projecting outward. Similarly, a boundary portion between first central portion 11 and first left inclined portion 13 may be curved as convexly projecting outward.

[0132] First upper surface 103 includes first right upper surface portion 16 and first left upper surface portion 17. First right upper surface portion 16 is located between first right inclined portion 12 and first right side surface 105. First right upper surface portion 16 is contiguous to each of first right inclined portion 12 and first right side surface 105. A boundary portion between first right upper surface portion 16 and first right side surface 105 may be curved as convexly projecting outward. First right inclined portion 12 is a portion inclined from first central portion 11 toward first right upper surface portion 16.

[0133] Similarly, first left upper surface portion 17 is located between first left inclined portion 13 and first left side surface 106. First left upper surface portion 17 is contiguous to each of first left inclined portion 13 and first left side surface 106. A boundary portion between first left upper surface portion 17 and first left side surface 106 may be curved as convexly projecting outward. First left inclined portion 13 is a portion inclined from first central portion 11 toward first left upper surface portion 17.

[0134] First device 100 further includes a first input portion 1, second input portion 2, a fifth input portion 5, and sixth input portion 6. First input portion 1 is arranged at first projecting portion 10. Specifically, first input portion 1 is provided at a portion of first projecting portion 10 at the first right side surface 105a side. When first display 110 is viewed in the plan view, first input portion 1 is provided closer to first right side surface 105 relative to a center of first upper surface 103. First input portion 1 is provided at first right inclined portion 12. First input portion 1 may be provided at a boundary portion between first central portion 11 and first right inclined portion 12.

[0135] Second input portion 2 is provided at first upper surface 103 other than first projecting portion 10. Specifically, second input portion 2 is provided at first right upper surface portion 16. Second input portion 2 may be provided at a boundary portion between first right upper surface portion 16 and first right side surface 105. Second input portion 2 is provided closer to first right side surface 105 relative to first input portion 1. In a direction from first lower surface 104 toward first upper surface 103, a lowermost portion of first input portion 1 may be located above an uppermost portion of second input portion 2.

[0136] Fifth input portion 5 is arranged at first projecting portion 10. Specifically, fifth input portion 5 is provided at a portion of first projecting portion 10 at the first left side surface 106 side. When first display 110 is viewed in the plan view, fifth input portion 5 is provided closer to first left side surface 106 relative to the center of first upper surface 103. Fifth input portion 5 is provided at first left inclined portion 13. Fifth input portion 5 may be provided at a boundary portion between first central portion 11 and first left inclined portion 13.

[0137] Sixth input portion 6 is provided at first upper surface 103 other than first projecting portion 10. Specifically, sixth input portion 6 is provided at first left upper surface portion 17. Sixth input portion 6 may be provided at a boundary portion between first left upper surface portion 17 and first left side surface 106. Sixth input portion 6 is provided closer to first left side surface 106 relative to fifth input portion 5. In the direction from first lower surface 104 toward first upper surface 103, a lowermost portion of fifth input portion 5 may be located above an uppermost portion of sixth input portion 6.

[0138] As shown in FIG. 20, when first display 110 is viewed in the plan view, first input portion 1 may be located closer to first right side surface 105 relative to first connection portion 120. Similarly, when first display 110 is viewed in the plan view, fifth input portion 5 may be located closer to first left side surface 106 relative to first connection portion 120.

[0139] FIG. 21 is a schematic cross-sectional view showing a configuration of fifth input portion 5 and sixth input portion 6. The cross-section shown in FIG. 21 is in parallel to first front surface 101. First device 100 includes a support member 40, a shaft receiving portion 41, an elastic body 44, a hole 42, and a detection portion 43. Shaft receiving portion 41 and hole 42 are recesses provided in support member 40. Elastic body 44 is, for example, a coil spring. Elastic body 44 is arranged in hole 42. Detection portion 43 is fixed to support member 40. Support member 40, shaft receiving portion 41, elastic body 44, hole 42, and detection portion 43 are arranged inside first housing 165.

[0140] Fifth input portion 5 includes a pressed portion 53, a movable shaft 51, a stopper portion 52, a second protruding portion 54, and a third protruding portion 55. Pressed portion 53 is a portion to be pushed by the user. Movable shaft 51 is arranged in shaft receiving portion 41. Movable shaft 51 is contiguous to pressed portion 53. Movable shaft 51 is in contact with an inner wall of first housing 165. Stopper portion 52 is contiguous to pressed portion 53. Stopper portion 52 is located opposite to movable shaft 51 with respect to pressed portion 53. Stopper portion 52 is in contact with the inner wall of first housing 165.

[0141] Second protruding portion 54 is contiguous to pressed portion 53. Second protruding portion 54 is arranged as being in contact with detection portion 43. Third protruding portion 55 is contiguous to pressed portion 53. Third protruding portion 55 is attached to elastic body 44. First projecting portion 10 of first housing 165 is provided with a fifth opening portion 70. Pressed portion 53 is exposed to the outside of first housing 165 through fifth opening portion 70.

[0142] When the user pushes pressed portion 53 with his/her finger, second protruding portion 54 moves to the inside of first housing 165. As second protruding portion 54 applies a pressure to detection portion 43, input by the user is detected. As the user's finger moves away from pressed

portion 53, fifth input portion 5 returns to an original position owing to repulsive force of elastic body 44. Each of movable shaft 51 and stopper portion 52 is in contact with the inner wall of first housing 165.

[0143] Though the configuration of fifth input portion 5 is described above, other input portions (specifically, first input portion 1, second input portion 2, a third input portion 3, a fourth input portion 4, sixth input portion 6, a seventh input portion 7, and an eighth input portion 8) are substantially the same in configuration to fifth input portion 5. Therefore, description of the configuration of other input portions will not be provided.

[0144] FIG. 22 is a schematic front view showing a configuration of second device 200 of electronic device 300 according to the third embodiment. As shown in FIG. 22, when second display 210 is viewed in the plan view, second housing 265 surrounds second display 210. Second housing 265 includes a second projecting portion 20. Second projecting portion 20 extends in the outward direction of second display 210. Second connection portion 220 may be provided at a rear side of second projecting portion 20.

[0145] Second projecting portion 20 is provided at second upper surface 203. Second projecting portion 20 may define a part of second front surface 201. Similarly, second projecting portion 20 may define a part of second rear surface 202. Second projecting portion 20 is distant from each of second right side surface 205, second left side surface 206, and second lower surface 204.

[0146] Second projecting portion 20 includes a second right inclined portion 22, a second left inclined portion 23, a second central portion 21, and a second front surface portion 24. Second right inclined portion 22 is located at a second right side surface 205 side of second projecting portion 20. Second left inclined portion 23 is located at a second left side surface 206 side of second projecting portion 20. Second central portion 21 is located between second right inclined portion 22 and second left inclined portion 23. Second central portion 21 is contiguous to each of second right inclined portion 22 and second left inclined portion 23. Second front surface portion 24 is contiguous to each of second right inclined portion 22, second left inclined portion 23, and second central portion 21. Second front surface portion 24 defines a part of second front surface 201.

[0147] As shown in FIG. 22, when second display 210 is viewed in the plan view, second central portion 21 is substantially in parallel to second lower surface 204. Each of second right inclined portion 22 and second left inclined portion 23 is inclined with respect to second central portion 21. A boundary portion between second central portion 21 and second right inclined portion 22 may be curved as convexly projecting outward. Similarly, a boundary portion between second central portion 21 and second left inclined portion 23 may be curved as convexly projecting outward.

[0148] Second upper surface 203 includes a second right upper surface portion 26 and a second left upper surface portion 27. Second right upper surface portion 26 is located between second right inclined portion 22 and second right side surface 205. Second right upper surface portion 26 is contiguous to each of second right inclined portion 22 and second right side surface 205. A boundary portion between second right upper surface portion 26 and second right side surface 205 may be curved as convexly projecting outward.

[0149] Similarly, second left upper surface portion 27 is located between second left inclined portion 23 and second

left side surface 206. Second left upper surface portion 27 is contiguous to each of second left inclined portion 23 and second left side surface 206. A boundary portion between second left upper surface portion 27 and second left side surface 206 may be curved as convexly projecting outward.

[0150] Second device 200 further includes third input portion 3, fourth input portion 4, seventh input portion 7, and eighth input portion 8. Third input portion 3 is arranged at second projecting portion 20. Specifically, third input portion 3 is provided at a portion of second projecting portion 20 at the second right side surface 205 side. When second display 210 is viewed in the plan view, third input portion 3 is provided closer to second right side surface 205 relative to a center of second upper surface 203. Third input portion 3 is provided at second right inclined portion 22. Third input portion 3 may be provided at a boundary portion between second central portion 21 and second right inclined portion 22.

[0151] Fourth input portion 4 is provided at second upper surface 203 other than second projecting portion 20. Specifically, fourth input portion 4 is provided at second right upper surface portion 26. Fourth input portion 4 may be provided at a boundary portion between second right upper surface portion 26 and second right side surface 205. Fourth input portion 4 is provided closer to second right side surface 205 relative to third input portion 3. In a direction from second lower surface 204 toward second upper surface 203, a lowermost portion of third input portion 3 may be located above an uppermost portion of fourth input portion 4.

[0152] Seventh input portion 7 is arranged at second projecting portion 20. Specifically, seventh input portion 7 is provided at a portion of second projecting portion 20 at the second left side surface 206 side. When second display 210 is viewed in the plan view, seventh input portion 7 is provided closer to second left side surface 206 relative to the center of second upper surface 203. Seventh input portion 7 is provided at second left inclined portion 23. Seventh input portion 7 may be provided at a boundary portion between second central portion 21 and second left inclined portion 23.

[0153] Eighth input portion 8 is provided at second upper surface 203 other than second projecting portion 20. Specifically, eighth input portion 8 is provided at second left upper surface portion 27. Eighth input portion 8 may be provided at a boundary portion between second left upper surface portion 27 and second left side surface 206. Eighth input portion 8 is provided closer to second left side surface 206 relative to seventh input portion 7. In the direction from second lower surface 204 toward second upper surface 203, a lowermost portion of seventh input portion 7 may be located above an uppermost portion of eighth input portion 8.

[0154] As shown in FIG. 22, a front surface (second front surface portion 24) of second projecting portion 20 may be provided with a functional component 29. Functional component 29 may be, for example, a camera lens or an input portion. Second front surface 201 other than second projecting portion 20 may be provided with an operation portion (not shown). For example, second front surface 201 other than second projecting portion 20 may be provided with a physical stick. According to such a configuration, a space at the front surface (second front surface portion 24) of second projecting portion 20 can effectively be made use of.

[0155] FIG. 23 is a schematic cross-sectional view along the line XXIII-XXIII in FIG. 22. As shown in FIG. 23, second display 210 includes a touch panel 32 and a touch panel cover 31. Touch panel 32 is a panel on which a touch operation can be performed. Touch panel 32 performs a function to detect touch. Touch panel cover 31 covers touch panel 32. Touch panel cover 31 can receive an input operation to touch panel 32. Touch panel cover 31 itself does not perform a function to detect touch. When the user touches touch panel cover 31, the input operation to touch panel 32 therethrough can be performed.

[0156] Second display 210 includes a first tactile feel presentation portion 81 and a second tactile feel presentation portion 82. First tactile feel presentation portion 81 is implemented, for example, by touch panel cover 31. Touch panel cover 31 is provided with an opening 30. Second tactile feel presentation portion 82 is, for example, a portion where touch panel 32 is exposed through opening 30. Second tactile feel presentation portion 82 is a part of touch panel 32. Second tactile feel presentation portion 82 may be used as a virtual button or a virtual stick.

[0157] Second tactile feel presentation portion 82 may be a part of touch panel cover 31. In this case, second tactile feel presentation portion 82 may be different from first tactile feel presentation portion 81 in sense of touch. Second tactile feel presentation portion 82 may be a portion of a surface of touch panel cover 31 that is roughened. In this case, first tactile feel presentation portion 81 is a portion of the surface of touch panel cover 31 that is not roughened. Second tactile feel presentation portion 82 may be a portion where vibration is generated as the user touches the same. In this case, first tactile feel presentation portion 81 is a portion where vibration is not generated even when the user touches the same.

[0158] FIG. 24 is a schematic perspective view showing a configuration of electronic device 300 according to the third embodiment in the first connection state. As shown in FIG. 24, in the first connection state, second rear surface 202 is arranged to cover first display 110. First projecting portion 10 is arranged as being superimposed on second projecting portion 20. Specifically, the front surface of first projecting portion 10 is superimposed on the rear surface of second projecting portion 20. First connection portion 120 provided at the front surface of first projecting portion 10 is connected, in first direction A1, to second connection portion 220 provided at the rear surface of second projecting portion 20.

[0159] In the first connection state, in the direction perpendicular to first front surface 101, first input portion 1 and third input portion 3 are arranged as being aligned and second input portion 2 and fourth input portion 4 are arranged as being aligned. From another point of view, along the direction from first front surface 101 toward second rear surface 202, first input portion 1 and third input portion 3 are arranged as being aligned and second input portion 2 and fourth input portion 4 are arranged as being aligned.

[0160] In the first connection state, in the direction perpendicular to first front surface 101, fifth input portion 5 and seventh input portion 7 are arranged as being aligned and sixth input portion 6 and eighth input portion 8 are arranged as being aligned. From another point of view, along the direction from first front surface 101 toward second rear surface 202, fifth input portion 5 and seventh input portion

7 are arranged as being aligned and sixth input portion 6 and eighth input portion 8 are arranged as being aligned.

[0161] FIG. 25 is a schematic perspective view showing a configuration of electronic device 300 according to the third embodiment in the second connection state. In the second connection state, second rear surface 202 is arranged not to cover first display 110. First projecting portion 10 is arranged as being superimposed on second projecting portion 20. Specifically, the front surface of first projecting portion 10 is superimposed on the rear surface of second projecting portion 20. First connection portion 120 provided at the front surface of first projecting portion 10 is connected, in second direction A2, to second connection portion 220 provided at the rear surface of second projecting portion 20.

[0162] In the second connection state, first device 100 and second device 200 are connected such that first projecting portion 10 is superimposed on second projecting portion 20, and hence first input portion 1, second input portion 2, fifth input portion 5, and sixth input portion 6 are structured to be distant from second device 200. Therefore, when each of first input portion 1, second input portion 2, fifth input portion 5, and sixth input portion 6 is operated, the user's finger is less likely to interfere with third input portion 3, fourth input portion 4, seventh input portion 7, and eighth input portion 8 or second projecting portion 20 of second device 200. Operability of electronic device 300 is thus improved.

[0163] Electronic device 300 according to a modification of the third embodiment does not have to include first input portion 1 to eighth input portion 8. Specifically, first device 100 does not have to include each of first input portion 1, second input portion 2, fifth input portion 5, and sixth input portion 6, and second device 200 does not have to include each of third input portion 3, fourth input portion 4, seventh input portion 7, and eighth input portion 8.

[0164] Though the configuration in which second rear surface 202 covers first display 110 in the first connection state and second rear surface 202 does not cover first display 110 in the second connection state is disclosed, in the first connection state in which second connection portion 220 is connected to first connection portion 120 in first direction A1, second rear surface 202 may be arranged to cover input portion 121, and in the second connection state in which second connection portion 220 is connected to first connection portion 120 in second direction A2, second rear surface 202 may be arranged not to cover input portion 121.

[0165] First input portion 1 may be configured to extend to first right upper surface portion 16. Third input portion 3 may be configured to extend to second right upper surface portion 26. Fifth input portion 5 may be configured to extend to first left upper surface portion 17. Seventh input portion 7 may be configured to extend to second left upper surface portion 27.

[0166] First input portion 1 and second input portion 2 may integrally be formed. A function as a button may be provided to a portion of each of first input portion 1 and second input portion 2. Third input portion 3 and fourth input portion 4 may integrally be formed. A function as a button may be provided to a portion of each of third input portion 3 and fourth input portion 4. Fifth input portion 5 and sixth input portion 6 may integrally be formed. A function as a button may be provided to a portion of each of fifth input portion 5 and sixth input portion 6. Seventh input portion 7

and eighth input portion 8 may integrally be formed. A function as a button may be provided to a portion of each of seventh input portion 7 and eighth input portion 8.

[0167] Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

What is claimed is:

1. An electronic device comprising:

a first device; and

a second device, wherein

the first device and the second device are removably attachable to each other,

the first device comprises

a first front surface, and

a first display and a first connection portion arranged at the first front surface,

the second device comprises

a second front surface,

a second display arranged at the second front surface,

a second rear surface located opposite to the second front surface, and

a second connection portion arranged at the second rear surface,

the second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction,

in a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display, and

in a second connection state in which the second connection portion is connected to the first connection portion in the second direction, the second rear surface is arranged not to cover the first display.

2. The electronic device according to claim 1, wherein the first device comprises

a first substrate arranged inside the first device, and

one first antenna or at least two first antennas provided at the first substrate,

the second device comprises

a second substrate arranged inside the second device, and

one second antenna or at least two second antennas provided at the second substrate,

in each of the first connection state and the second connection state, when the first display is viewed in a plan view, the first antenna and the second antenna are arranged at positions superimposed on each other, and the first device and the second device are configured to perform first wireless communication through electromagnetic field coupling between the first antenna and the second antenna.

3. The electronic device according to claim 2, wherein

in each of the first connection state and the second connection state, at least a part of the first antenna is arranged at a position superimposed on the second antenna.

4. The electronic device according to claim 3, wherein

one of the first antenna or the second antenna comprises a first antenna portion,

the other of the first antenna or the second antenna comprises a second antenna portion and a third antenna portion,

in the first connection state, the first antenna portion is arranged as being superimposed on the second antenna portion, and

in the second connection state, the first antenna portion is arranged as being superimposed on the third antenna portion.

5. The electronic device according to claim 4, wherein

the first antenna comprises the first antenna portion,

the second antenna comprises the second antenna portion and the third antenna portion,

when the first front surface is viewed in the plan view, the first antenna portion is arranged at a position adjacent to the first connection portion, and

when the second rear surface is viewed in the plan view, the second connection portion is arranged as lying between the second antenna portion and the third antenna portion.

6. The electronic device according to claim 1, wherein

the first device comprises

a first substrate arranged inside the first device, and

one first antenna or at least two first antennas provided at the first substrate,

the second device comprises

a second substrate arranged inside the second device, and

one second antenna or at least two second antennas provided at the second substrate,

one of the first antenna or the second antenna comprises a first antenna portion,

the other of the first antenna or the second antenna comprises a second antenna portion and a third antenna portion,

in the first connection state, the second antenna portion is arranged as being superimposed on the first antenna portion and the third antenna portion is arranged as not being superimposed on the first antenna, and

in the second connection state, the third antenna portion is arranged as being superimposed on the first antenna portion and the second antenna portion is arranged as not being superimposed on the first antenna.

7. The electronic device according to claim 6, wherein

when the first front surface is viewed in a plan view, the first antenna portion is arranged at a position adjacent to the first connection portion, and

when the second rear surface is viewed in the plan view, the second connection portion is arranged as lying between the second antenna portion and the third antenna portion.

8. The electronic device according to claim 2, wherein

in the second connection state, the first device and the second device are connected with the second display being inclined with respect to the first display.

9. The electronic device according to claim 2, wherein

in the first connection state, the first device and the second device are connected with the second display being in parallel to the first display.

10. The electronic device according to claim 8, wherein the first connection portion comprises a first inclined surface inclined with respect to the first front surface, and

the second connection portion comprises a second inclined surface inclined with respect to the second rear surface.

11. The electronic device according to claim **10**, wherein when an angle between the first antenna and the second antenna in a cross-section perpendicular to each of the first front surface and the second rear surface in the first connection state is defined as a first angle, an angle between the first antenna and the second antenna in the cross-section perpendicular to each of the first front surface and the second rear surface in the second connection state is defined as a second angle, an angle between the first front surface and the first inclined surface in the cross-section perpendicular to the first front surface is defined as a third angle, and an angle between the second rear surface and the second inclined surface in the cross-section perpendicular to the second rear surface is defined as a fourth angle, each of the first angle and the second angle is smaller than a total of the third angle and the fourth angle.

12. The electronic device according to claim **2**, wherein when the first antenna and the second antenna are electromagnetically coupled to each other, the first device and the second device transmit and receive a signal unidirectionally or bidirectionally, the first connection portion comprises a first contact for electric power supply, the second connection portion comprises a second contact for electric power supply, and in each of the first connection state and the second connection state, the first contact and the second contact are connected.

13. The electronic device according to claim **2**, wherein the first device and the second device are configured to perform second wireless communication while the first device and the second device are separate.

14. The electronic device according to claim **13**, wherein the first wireless communication in the first connection state and the second connection state is performed through electromagnetic field coupling between the first antenna and the second antenna, the second wireless communication while the first device and the second device are separate is performed through electromagnetic field coupling between antennas different from each of the first antenna and the second antenna, and a frequency band of a signal to be used in the first wireless communication is higher than a frequency band of a signal to be used in the second wireless communication.

15. The electronic device according to claim **1**, wherein the first connection portion comprises
a first receiving groove, and
a second receiving groove located opposite to the first receiving groove,
the second connection portion comprises
a movable hook, and
a fixed hook located opposite to the movable hook,
in the first connection state, the movable hook is not arranged in the first receiving groove and the fixed hook is arranged in the second receiving groove, and
in the second connection state, the movable hook is arranged in the second receiving groove and the fixed hook is arranged in the first receiving groove.

16. The electronic device according to claim **1**, wherein the first connection portion comprises a first electrical connection portion and a first mechanical connection portion,

the first electrical connection portion is provided above or below the first display,

the first mechanical connection portion is provided to sandwich the first electrical connection portion from a longitudinal direction of the first display,

the second connection portion comprises a second electrical connection portion and a second mechanical connection portion,

the second electrical connection portion is provided above or below the second display,

the second mechanical connection portion is provided to sandwich the second electrical connection portion from a longitudinal direction of the second display,

the first electrical connection portion is to electrically be connected to the second electrical connection portion, and

the first mechanical connection portion is to mechanically be connected to the second mechanical connection portion.

17. The electronic device according to claim **16**, wherein the first electrical connection portion has terminals disposed in the longitudinal direction of the first display, and

the second electrical connection portion has a terminal disposed in the longitudinal direction of the second display.

18. The electronic device according to claim **1**, wherein the first device comprises an input portion configured to be operated as a stick or a button, and

the second display comprises a touch panel on which a touch input can be provided.

19. The electronic device according to claim **1**, wherein the first device comprises a first housing that surrounds the first display and comprises a first projecting portion that extends in an outward direction of the first display, the second device comprises a second housing that surrounds the second display and comprises a second projecting portion that extends in an outward direction of the second display,

the first connection portion is provided at a front side of the first projecting portion, and

the second connection portion is provided at a rear side of the second projecting portion.

20. The electronic device according to claim **19**, wherein the first device comprises

a first rear surface located opposite to the first front surface,

a first upper surface contiguous to the first front surface and the first rear surface,

a first lower surface located opposite to the first upper surface,

a first right side surface contiguous to each of the first front surface, the first rear surface, the first upper surface, and the first lower surface, and

a first left side surface located opposite to the first right side surface, the second device comprises

a second upper surface contiguous to the second front surface and the second rear surface,

a second lower surface located opposite to the second upper surface,

a second right side surface contiguous to each of the second front surface, the second rear surface, the second upper surface, and the second lower surface, and
a second left side surface located opposite to the second right side surface,
the first projecting portion is provided at the first upper surface,
the second projecting portion is provided at the second upper surface,
the first device comprises
a first input portion provided at a portion of the first projecting portion at a side of the first right side surface, and
a second input portion provided at the first upper surface other than the first projecting portion, and
the second device comprises
a third input portion provided at a portion of the second projecting portion at a side of the second right side surface, and
a fourth input portion provided at the second upper surface other than the second projecting portion.

21. The electronic device according to claim **20**, wherein in the first connection state, in a direction perpendicular to the first front surface, the first input portion and the third input portion are arranged as being aligned and the second input portion and the fourth input portion are arranged as being aligned.

22. The electronic device according to claim **19**, wherein at a surface of the second projecting portion, an input portion or a camera lens is provided.

23. The electronic device according to claim **1**, wherein the second display comprises

a touch panel on which a touch operation can be performed, and

a touch panel cover on which an input operation to the touch panel can be performed, the touch panel cover covering the touch panel, and

the second display comprises

a first tactile feel presentation portion configured with the touch panel cover, and

a second tactile feel presentation portion where the touch panel is exposed in an opening provided in the touch panel cover.

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